

Answer key

1.18.22 Graph Search: GATE IT 2007 | Question: 24

A depth-first search is performed on a directed acyclic graph. Let $d[u]$ denote the time at which vertex u is visited for the first time and $f[u]$ the time at which the DFS call to the vertex u terminates. Which of the following statements is always TRUE for all edges (u, v) in the graph ?

- A. $d[u] < d[v]$ B. $d[u] < f[v]$ 2
C. $f[u] < f[v]$ D. $f[u] > f[v]$

gateit-2007 algorithms graph-algorithms normal graph-search depth-first-search

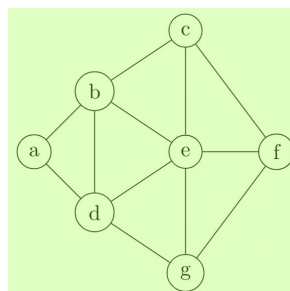
Answer key 3

1.18.23 Graph Search: GATE IT 2008 | Question: 47

Consider the following sequence of nodes for the undirected graph given below: 5

1. $a b e f d g c$ 6
2. $a b e f c g d$
3. $a d g e b c f$
4. $a d b c g e f$

A Depth First Search (DFS) is started at node a . The nodes are listed in the order they are first visited. Which of the above is/are possible output(s)? 7



- A. 1 and 3 only B. 2 and 3 only C. 2, 3 and 4 only D. 1, 2 and 3 only 9

gateit-2008 algorithms graph-algorithms normal graph-search depth-first-search 10

Answer key 11

1.19

Greedy Algorithms (5)

Practice Test: Test 1 (7Q) 12

1.19.1 Greedy Algorithms: GATE CSE 1999 | Question: 2.20

The minimum number of record movements required to merge five files A (with 10 records), B (with 20 records), C (with 15 records), D (with 5 records) and E (with 25 records) is: 13

- A. 165 B. 90 C. 75 D. 65 14

gate1999 algorithms normal greedy-algorithms 15

Answer key 16

1.19.2 Greedy Algorithms: GATE CSE 2003 | Question: 69

The following are the starting and ending times of activities A, B, C, D, E, F, G and H respectively in chronological order: " $a_s b_s c_s a_e d_s c_e e_s f_s b_e d_e g_s e_e f_e h_s g_e h_e$ ". Here, x_s denotes the starting time and x_e denotes the ending time of activity X . We need to schedule the activities in a set of rooms available to us. An activity can be scheduled in a room only if the room is reserved for the activity for its entire duration. What is the minimum number of rooms required?

A. 3 B. 4 C. 5 D. 6 ¹

gatecse-2003 algorithms normal greedy-algorithms ²

Answer key ³

1.19.3 Greedy Algorithms: GATE CSE 2005 | Question: 84a



We are given 9 tasks $T_1, T_2, \dots, T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i . Profit P_i is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

⁵

Are all tasks completed in the schedule that gives maximum profit? ⁶

- A. All tasks are completed B. T_1 and T_6 are left out ⁷
 C. T_1 and T_8 are left out D. T_4 and T_6 are left out

gatecse-2005 algorithms greedy-algorithms process-scheduling normal ⁸

Answer key ⁹

1.19.4 Greedy Algorithms: GATE CSE 2005 | Question: 84b



We are given 9 tasks $T_1, T_2, \dots, T_9$. The execution of each task requires one unit of time. We can execute one task at a time. Each task T_i has a profit P_i and a deadline d_i . Profit P_i is earned if the task is completed before the end of the d_i^{th} unit of time.

Task	T_1	T_2	T_3	T_4	T_5	T_6	T_7	T_8	T_9
Profit	15	20	30	18	18	10	23	16	25
Deadline	7	2	5	3	4	5	2	7	3

¹¹

What is the maximum profit earned? ¹²

- A. 147 B. 165 C. 167 D. 175 ¹³

gatecse-2005 algorithms greedy-algorithms process-scheduling normal ¹⁴

Answer key ¹⁵

1.19.5 Greedy Algorithms: GATE CSE 2018 | Question: 48



Consider the weights and values of items listed below. Note that there is only one unit of each item. ¹⁶

Item number	Weight (in Kgs)	Value (in rupees)
1	10	60
2	7	28
3	4	20
4	2	24

¹⁷

The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt} . A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy} .

The value of $V_{opt} - V_{greedy}$ is ¹⁹

gatecse-2018 algorithms greedy-algorithms numerical-answers two-marks

1.20.1 Hashing: GATE CSE 1989 | Question: 1-vii, ISRO2015-14



A hash table with ten buckets with one slot per bucket is shown in the following figure. The symbols S_1 to S_7 initially entered using a hashing function with linear probing. The maximum number of comparisons needed in searching an item that is not present is

0	S_7
1	S_1
2	
3	S_4
4	S_2
5	
6	S_5
7	
8	S_6
9	S_3

2

- A. 4 B. 5 C. 6 D. 3

hashing isro2015 gate1989 algorithms normal

4

Answer key

5

1.20.2 Hashing: GATE CSE 1990 | Question: 13b



Consider a hash table with chaining scheme for overflow handling:

- What is the worst-case timing complexity of inserting n elements into such a table?
- For what type of instance does this hashing scheme take the worst-case time for insertion?

7

gate1990 hashing algorithms descriptive

Answer key

1.20.3 Hashing: GATE CSE 2021 | Set 1 | Question: 47

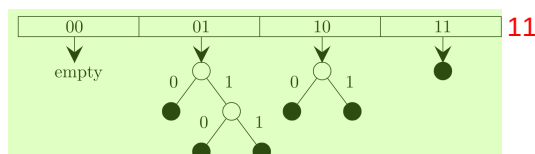


Consider a *dynamic* hashing approach for 4-bit integer keys:

- There is a main hash table of size 4.
- The 2 least significant bits of a key is used to index into the main hash table.
- Initially, the main hash table entries are empty.
- Thereafter, when more keys are hashed into it, to resolve collisions, the set of all keys corresponding to a main hash table entry is organized as a binary tree that grows on demand.
- First, the 3rd least significant bit is used to divide the keys into left and right subtrees.
- To resolve more collisions, each node of the binary tree is further sub-divided into left and right subtrees based on the 4th least significant bit.
- A split is done only if it is needed, i.e., only when there is a collision.

9

Consider the following state of the hash table:



11

Which of the following sequences of key insertions can cause the above state of the hash table (assume the keys are in decimal notation)?

12

- A. 5, 9, 4, 13, 10, 7 B. 9, 5, 10, 6, 7, 1

13

C. 10,9,6,7,5,13

D. 9,5,13,6,10,14¹

gatecse-2021-set1 multiple-selects algorithms hashing two-marks²

Answer key

1.20.4 Hashing: GATE CSE 2023 | Question: 10

An algorithm has to store several keys generated by an adversary in a hash table. The adversary is malicious who tries to maximize the number of collisions. Let k be the number of keys, m be the number of slots in the hash table, and $k > m$.

Which one of the following is the best hashing strategy to counteract the adversary?⁴

- A. Division method, i.e., use the hash function $h(k) = k \bmod m$.
- B. Multiplication method, i.e., use the hash function $h(k) = \lfloor m(kA - \lfloor kA \rfloor) \rfloor$, where A is a carefully chosen constant.
- C. Universal hashing method.
- D. If k is a prime number, use Division method. Otherwise, use Multiplication method.

gatecse-2023 algorithms hashing one-mark

Answer key

1.20.5 Hashing: GATE CSE 2026 | Set 2 | Question: 20

The keys 5, 28, 19, 15, 26, 33, 12, 17, 10 are inserted into a hash table using the hash function $h(k) = k \bmod 9$. The collisions are resolved by chaining. After all the keys are inserted, the length of the longest chain is _____. (answer in integer)

gatecse-2026-set2 algorithms hashing numerical-answers one-mark⁷

Answer key

1.20.6 Hashing: GATE IT 2005 | Question: 16

A hash table contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is $\text{key} \% 10$. If the values 43, 165, 62, 123, 142 are inserted in the table, in what location would the key value 142 be inserted?

- A. 2
- B. 3
- C. 4
- D. 6⁹

gateit-2005 algorithms hashing easy¹⁰

Answer key

1.21 Heap Sort (2)

1.21.1 Heap Sort: GATE CSE 2013 | Question: 30

The number of elements that can be sorted in $\Theta(\log n)$ time using heap sort is¹¹

- A. $\Theta(1)$
- B. $\Theta(\sqrt{\log n})$ ¹²
- C. $\Theta(\frac{\log n}{\log \log n})$
- D. $\Theta(\log n)$

gatecse-2013 algorithms sorting normal heap-sort¹³

Answer key

1.21.2 Heap Sort: GATE CSE 2024 | Set 1 | Question: 31

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

- A. 82, 90, 101
- B. 82, 11, 93
- C. 131, 11, 93
- D. 131, 111, 90¹⁶

gatecse-2024-set1 algorithms heap-sort sorting two-marks¹⁷

Answer key

1.22 Huffman Code (6)

1.22.1 Huffman Code: GATE CSE 1989 | Question: 13a



A language uses an alphabet of six letters, $\{a, b, c, d, e, f\}$. The relative frequency of use of each letter of the alphabet in the language is as given below:

LETTER	RELATIVE FREQUENCY OF USE
a	0.19
b	0.05
c	0.17
d	0.08
e	0.40
f	0.11

Design a prefix binary code for the language which would minimize the average length of the encoded words of the language.

descriptive gate1989 algorithms huffman-code 4

Answer key

1.22.2 Huffman Code: GATE CSE 2007 | Question: 76



Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

Which of the following is the Huffman code for the letter a, b, c, d, e, f ?

- A. 0, 10, 110, 1110, 11110, 11111 B. 11, 10, 011, 010, 001, 000
- C. 11, 10, 01, 001, 0001, 0000 D. 110, 100, 010, 000, 001, 111

gatecse-2007 algorithms greedy-algorithms normal huffman-code 7

Answer key

1.22.3 Huffman Code: GATE CSE 2007 | Question: 77



Suppose the letters a, b, c, d, e, f have probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}$, respectively.

What is the average length of the Huffman code for the letters a, b, c, d, e, f ?

- A. 3 B. 2.1875 C. 2.25 D. 1.9375

gatecse-2007 algorithms greedy-algorithms normal huffman-code 11

Answer key

1.22.4 Huffman Code: GATE CSE 2017 | Set 2 | Question: 50



A message is made up entirely of characters from the set $X = \{P, Q, R, S, T\}$. The table of probabilities for each of the characters is shown below:

Character	Probability
P	0.22
Q	0.34
R	0.17
S	0.19
T	0.08
Total	1.00

If a message of 100 characters over X is encoded using Huffman coding, then the expected length of the encoded

message in bits is _____.1

gatecse-2017-set2 huffman-code numerical-answers algorithms 2

Answer key

1.22.5 Huffman Code: GATE CSE 2021 | Set 2 | Question: 26

Consider the string `abbccddeee`. Each letter in the string must be assigned a binary code satisfying the following properties:

1. For any two letters, the code assigned to one letter must not be a prefix of the code assigned to the other letter. 4
2. For any two letters of the same frequency, the letter which occurs earlier in the dictionary order is assigned a code whose length is at most the length of the code assigned to the other letter.

Among the set of all binary code assignments which satisfy the above two properties, what is the minimum length of the encoded string? 5

- A. 21 B. 23 C. 25 D. 30 6

gatecse-2021-set2 algorithms huffman-code two-marks 7

Answer key

1.22.6 Huffman Code: GATE IT 2006 | Question: 48

The characters a to h have the set of frequencies based on the first 8 Fibonacci numbers as follows 8
 $a : 1, b : 1, c : 2, d : 3, e : 5, f : 8, g : 13, h : 21$

A Huffman code is used to represent the characters. What is the sequence of characters corresponding to the following code? 9

110111100111010 10

- A. *fdheg* B. *ecgdf* C. *dchfg* D. *fehdg* 11

gateit-2006 algorithms greedy-algorithms normal huffman-code 12

Answer key 13

1.23 Identify Function (38)

 **Practice Tests:** Test 1 (15Q) Test 2 (15Q) Test 3 (9Q) 14

1.23.1 Identify Function: GATE CSE 1989 | Question: 8a

What is the output produced by the following program, when the input is "HTGATE" 15

```
Function what (s:string): string;
var n:integer;
begin
    n = s.length
    if n <= 1
    then what := s
    else what :=contact (what (substring (s, 2, n)), s.C [1])
end;
```

Note 17

- i. type string=record
 length:integer;
 C:array[1..100] of char
end
- ii. Substring (s, i, j): this yields the string made up of the i^{th} through j^{th} characters in s ; for appropriately defined in i and j .
- iii. Contact (s_1, s_2): this function yields a string of length s_1 length + s_2 - length obtained by concatenating s_1 with s_2 such that s_1 precedes s_2 . 18

gate1989 descriptive algorithms identify-function

Answer key

1.23.2 Identify Function: GATE CSE 1990 | Question: 11b



The following program computes values of a mathematical function $f(x)$. Determine the form of $f(x)$.¹

```
main ()
{
    int m, n; float x, y, t;
    scanf ("%f%d", &x, &n);
    t = 1; y = 0; m = 1;
    do
    {
        t *= (-x/m);
        y += t;
    } while (m++ < n);
    printf ("The value of y is %f", y);
}
```

2

gate1990 descriptive algorithms identify-function 3

Answer key

1.23.3 Identify Function: GATE CSE 1991 | Question: 03-viii



Consider the following Pascal function:⁴

```
Function X(M:integer):integer;
Var i:integer;
Begin
    i := 0;
    while i*i < M
    do i := i+1
    X := i
end
```

5

The function call $X(N)$, if N is positive, will return⁶

- A. $\lfloor \sqrt{N} \rfloor$
- B. $\lfloor \sqrt{N} \rfloor + 1$ ⁷
- C. $\lceil \sqrt{N} \rceil$
- D. $\lceil \sqrt{N} \rceil + 1$
- E. None of the above

gate1991 algorithms easy identify-function multiple-selects 8

Answer key

1.23.4 Identify Function: GATE CSE 1993 | Question: 7.4



What does the following code do?⁹

```
var a, b: integer;
begin
    a:=a+b;
    b:=a-b;
    a:=a-b;
end;
```

10

- A. exchanges a and b
- B. doubles a and stores in b ¹¹
- C. doubles b and stores in a
- D. leaves a and b unchanged
- E. none of the above

gate1993 algorithms identify-function easy 12

Answer key

1.23.5 Identify Function: GATE CSE 1994 | Question: 6



What function of x, n is computed by this program?

```
Function what(x, n:integer): integer;
Var
    value : integer
begin
    value := 1
    if n > 0 then
```

```

begin
  if n mod 2 = 1 then
    value := value * x;
    value := value * what(x*x, n div 2);
  end;
  what := value;
end;

```

1

gate1994 algorithms identify-function normal descriptive

Answer key

1.23.6 Identify Function: GATE CSE 1995 | Question: 1.4



In the following Pascal program segment, what is the value of X after the execution of the program segment?

```

X := -10; Y := 20;
If X > Y then if X < 0 then X := abs(X) else X := 2*X;

```

3

- A. 10 B. -20 C. -10 D. None

4

gate1995 algorithms identify-function easy

5

Answer key

1.23.7 Identify Function: GATE CSE 1995 | Question: 2.3



Assume that X and Y are non-zero positive integers. What does the following Pascal program segment do?

```

while X <> Y do
if X > Y then
  X := X - Y
else
  Y := Y - X;
write(X);

```

7

- A. Computes the LCM of two numbers B. Divides the larger number by the smaller number
C. Computes the GCD of two numbers D. None of the above

8

gate1995 algorithms identify-function normal

9

Answer key

1.23.8 Identify Function: GATE CSE 1995 | Question: 4



- A. Consider the following Pascal function where A and B are non-zero positive integers. What is the value of $GET(3, 2)$?

10

```

function GET(A,B:integer): integer;
begin
  if B=0 then
    GET:= 1
  else if A < B then
    GET:= 0
  else
    GET:= GET(A-1, B) + GET(A-1, B-1)
end;

```

11

- B. The Pascal procedure given for computing the transpose of an $N \times N$, ($N > 1$) matrix A of integers has an error. Find the error and correct it. Assume that the following declaration are made in the main program

12

```

const
  MAXSIZE=20;
type
  INTARR=array [1..MAXSIZE,1..MAXSIZE] of integer;
Procedure TRANSPOSE (var A: INTARR; N : integer);
var
  I, J, TMP: integer;
begin
  for I:=1 to N - 1 do
    for J:=1 to N do
      begin
        TMP:= A[I, J];
        A[I, J]:= A[J, I];

```

13


```

    A[J, I]:= TMP
  end
end;

```

gate1995 algorithms identify-function normal descriptive

Answer key

1.23.9 Identify Function: GATE CSE 1998 | Question: 2.12



What value would the following function return for the input $x = 95$? **1**

```

Function fun (x:integer):integer;
Begin
  If x > 100 then fun = x - 10
  Else fun = fun(fun (x+11))
End;

```

- A. 89 B. 90 C. 91 D. 92 **3**

gate1998 algorithms recursion identify-function normal **4**

Answer key

1.23.10 Identify Function: GATE CSE 1999 | Question: 2.24



Consider the following C function definition **5**

```

int Trial (int a, int b, int c)
{
  if ((a>=b) && (c<b)) return b;
  else if (a>=b) return Trial(a, c, b);
  else return Trial(b, a, c);
}

```

The functional Trial: **7**

- A. Finds the maximum of a , b , and c B. Finds the minimum of a , b , and c **8**
 C. Finds the middle number of a , b , c D. None of the above

gate1999 algorithms identify-function normal **9**

Answer key

1.23.11 Identify Function: GATE CSE 2000 | Question: 2.15



Suppose you are given an array $s[1....n]$ and a procedure reverse(s, i, j) which reverses the order of elements in s between positions i and j (both inclusive). What does the following sequence do, where $1 \leq k \leq n$: **10**

```

reverse (s, 1, k);
reverse (s, k+1, n);
reverse (s, 1, n);

```

11

- A. Rotates s left by k positions B. Leaves s unchanged **12**
 C. Reverses all elements of s D. None of the above

gatecse-2000 algorithms normal identify-function **13**

Answer key

1.23.12 Identify Function: GATE CSE 2003 | Question: 1



Consider the following C function.

For large values of y , the return value of the function f best approximates

```

float f,(float x, int y) {
  float p, s; int i;
  for (s=1,p=1,i=1; i<y; i++) {
    p *= x/i;
    s += p;
  }
  return s;
}

```

- A. x^y B. e^x C. $\ln(1+x)$ D. x^x 1

gatecse-2003 algorithms identify-function normal

Answer key

1.23.13 Identify Function: GATE CSE 2003 | Question: 88



In the following C program fragment, j , k , n and `TwoLog_n` are integer variables, and A is an array of integers. The variable n is initialized to an integer ≥ 3 , and `TwoLog_n` is initialized to the value of $2^{\lceil \log_2(n) \rceil}$

```
for (k = 3; k <= n; k++)
    A[k] = 0;
for (k = 2; k <= TwoLog_n; k++)
    for (j = k+1; j <= n; j++)
        A[j] = A[j] || (j%k);
for (j = 3; j <= n; j++)
    if (!A[j]) printf("%d", j);
```

The set of numbers printed by this program fragment is 4

- A. $\{m \mid m \leq n, (\exists i) [m = i!]\}$ B. $\{m \mid m \leq n, (\exists i) [m = i^2]\}$ 5
C. $\{m \mid m \leq n, m \text{ is prime}\}$ D. $\{\}$

gatecse-2003 algorithms identify-function normal 6

Answer key

1.23.14 Identify Function: GATE CSE 2004 | Question: 41



Consider the following C program 7

```
main()
{
    int x, y, m, n;
    scanf("%d %d", &x, &y);
    /* Assume x>0 and y>0 */
    m = x; n = y;
    while (m != n)
    {
        if (m > n)
            m = m - n;
        else
            n = n - m;
    }
    printf("%d", n);
}
```

The program computes 9

- A. $x + y$ using repeated subtraction B. $x \bmod y$ using repeated subtraction 10
C. the greatest common divisor of x and y D. the least common multiple of x and y

gatecse-2004 algorithms normal identify-function 11

Answer key

1.23.15 Identify Function: GATE CSE 2004 | Question: 42



What does the following algorithm approximate? (Assume $m > 1, \epsilon > 0$).

```
x = m;
y = 1;
While (x-y > ε)
{
    x = (x+y)/2;
    y = m/x;
}
print(x);
```

- A. $\log m$ B. m^2 C. $m^{\frac{1}{2}}$ D. $m^{\frac{1}{3}}$

gatecse-2004 algorithms identify-function normal

1.23.16 Identify Function: GATE CSE 2005 | Question: 31



Consider the following C-program: 1

```
void foo (int n, int sum) { 2
    int k = 0, j = 0;
    if (n == 0) return;
    k = n % 10; j = n/10;
    sum = sum + k;
    foo (j, sum);
    printf ("%d,",k);
}

int main() {
    int a = 2048, sum = 0;
    foo(a, sum);
    printf ("%d\n", sum);
}
```

What does the above program print? 3

- A. 8, 4, 0, 2, 14 B. 8, 4, 0, 2, 0 4
C. 2, 0, 4, 8, 14 D. 2, 0, 4, 8, 0

gatecse-2005 algorithms identify-function recursion normal 5

1.23.17 Identify Function: GATE CSE 2006 | Question: 50



A set X can be represented by an array $x[n]$ as follows: 6

$$x[i] = \begin{cases} 1 & \text{if } i \in X \\ 0 & \text{otherwise} \end{cases} 7$$

Consider the following algorithm in which x , y , and z are Boolean arrays of size n : 8

```
algorithm zzz(x[], y[], z[]) {
    int i;
    for(i=0; i<n; ++i)
        z[i] = (x[i] ^ ~y[i]) ^ (~x[i] ^ y[i]);
}
```

The set Z computed by the algorithm is: 10

- A. $(X \cup Y)$ B. $(X \cap Y)$ C. $(X - Y) \cap (Y - X)$ D. $(X - Y) \cup (Y - X)$ 11

gatecse-2006 algorithms identify-function normal 12

1.23.18 Identify Function: GATE CSE 2006 | Question: 53



Consider the following C-function in which $a[n]$ and $b[m]$ are two sorted integer arrays and $c[n+m]$ be another integer array, 13

```
void xyz(int a[], int b[], int c[]) { 14
    int i, j, k;
    i=j=k=0;
    while ((i<n) && (j<m))
        if (a[i] < b[j]) c[k++] = a[i++];
        else c[k++] = b[j++];
}
```

Which of the following condition(s) hold(s) after the termination of the while loop? 15

- i. $j < m, k = n + j - 1$ and $a[n-1] < b[j]$ if $i = n$ 16
ii. $i < n, k = m + i - 1$ and $b[m-1] \leq a[i]$ if $j = m$

- A. only (i) B. only (ii) 17
C. either (i) or (ii) but not both D. neither (i) nor (ii)

gatecse-2006 algorithms identify-function normal 18

Answer key

1.23.19 Identify Function: GATE CSE 2009 | Question: 18



Consider the program below: 1

```
#include <stdio.h>
int fun(int n, int *f_p) {
    int t, f;
    if (n <= 1) {
        *f_p = 1;
        return 1;
    }
    t = fun(n-1, f_p);
    f = t + *f_p;
    *f_p = t;
    return f;
}

int main() {
    int x = 15;
    printf("%d/n", fun(5, &x));
    return 0;
}
```

2

The value printed is: 3

- A. 6 B. 8 C. 14 D. 15 4

gatecse-2009 algorithms recursion identify-function normal

Answer key

1.23.20 Identify Function: GATE CSE 2010 | Question: 35



What is the value printed by the following C program? 5

```
#include<stdio.h>
int f(int *a, int n)
{
    if (n <= 0) return 0;
    else if (*a % 2 == 0) return *a+f(a+1, n-1);
    else return *a - f(a+1, n-1);
}

int main()
{
    int a[] = {12, 7, 13, 4, 11, 6};
    printf("%d", f(a, 6));
    return 0;
}
```

6

- A. -9 B. 5 C. 15 D. 19 7

gatecse-2010 algorithms recursion identify-function normal 8

Answer key

1.23.21 Identify Function: GATE CSE 2011 | Question: 48



Consider the following recursive C function that takes two arguments.

```
unsigned int foo(unsigned int n, unsigned int r) {
    if (n>0) return ((n%r) + foo(n/r, r));
    else return 0;
}
```

What is the return value of the function foo when it is called as foo(345, 10)?

- A. 345 B. 12 C. 5 D. 3

gatecse-2011 algorithms recursion identify-function normal

Answer key

1.23.22 Identify Function: GATE CSE 2011 | Question: 49



Consider the following recursive C function that takes two arguments. **1**

```
unsigned int foo(unsigned int n, unsigned int r) { 2
    if (n>0) return ((n%r) + foo(n/r, r));
    else return 0;
}
```

What is the return value of the function `foo` when it is called as `foo(513, 2)`? **3**

- A. 9 B. 8 C. 5 D. 2 **4**

gatecse-2011 algorithms recursion identify-function normal **5**

Answer key

1.23.23 Identify Function: GATE CSE 2013 | Question: 31



Consider the following function: **6**

```
int unknown(int n){ 7
    int i, j, k=0;
    for (i=n/2; i<=n; i++)
        for (j=2; j<=n; j=j*2)
            k = k + n/2;
    return (k);
}
```

The return value of the function is **8**

- A. $\Theta(n^2)$ B. $\Theta(n^2 \log n)$ **9**
 C. $\Theta(n^3)$ D. $\Theta(n^3 \log n)$

gatecse-2013 algorithms identify-function normal **10**

Answer key

1.23.24 Identify Function: GATE CSE 2014 | Set 1 | Question: 41



Consider the following C function in which **size** is the number of elements in the array **E**: **11**

```
int MyX(int *E, unsigned int size) 12
{
    int Y = 0;
    int Z;
    int i, j, k;

    for(i = 0; i < size; i++)
        Y = Y + E[i];

    for(i=0; i < size; i++)
        for(j = i; j < size; j++)
        {
            Z = 0;
            for(k = i; k <= j; k++)
                Z = Z + E[k];
            if(Z > Y)
                Y = Z;
        }
    return Y;
}
```

The value returned by the function **MyX** is the **13**

- A. maximum possible sum of elements in any sub-array of array **E**. **14**
 B. maximum element in any sub-array of array **E**.
 C. sum of the maximum elements in all possible sub-arrays of array **E**.
 D. the sum of all the elements in the array **E**.

gatecse-2014-set1 algorithms identify-function normal

Answer key

1.23.25 Identify Function: GATE CSE 2014 | Set 2 | Question: 10



Consider the function func shown below: 1

```
int func(int num) { 2
    int count = 0;
    while (num) {
        count++;
        num >>= 1;
    }
    return (count);
}
```

The value returned by func(435) is 3

gatescse-2014-set2 algorithms identify-function numerical-answers easy 4

Answer key

1.23.26 Identify Function: GATE CSE 2014 | Set 3 | Question: 10



Let A be the square matrix of size $n \times n$. Consider the following pseudocode. What is the expected output? 5

```
C=100;
for i=1 to n do
    for j=1 to n do
    {
        Temp = A[i][j]+C;
        A[i][j] = A[j][i];
        A[j][i] = Temp -C;
    }
for i=1 to n do
    for j=1 to n do
        output (A[i][j]);
```

- 6
- A. The matrix A itself
 - B. Transpose of the matrix A
 - C. Adding 100 to the upper diagonal elements and subtracting 100 from lower diagonal elements of A
 - D. None of the above

gatescse-2014-set3 algorithms identify-function easy

Answer key

1.23.27 Identify Function: GATE CSE 2015 | Set 1 | Question: 31



Consider the following C function. 8

```
int fun1 (int n) { 9
    int i, j, k, p, q = 0;
    for (i = 1; i < n; ++i)
    {
        p = 0;
        for (j = n; j > 1; j = j/2)
            ++p;
        for (k = 1; k < p; k = k * 2)
            ++q;
    }
    return q;
}
```

Which one of the following most closely approximates the return value of the function fun1? 10

- A. n^3
- B. $n(\log n)^2$
- C. $n \log n$
- D. $n \log(\log n)$ 11

gatescse-2015-set1 algorithms normal identify-function 12

Answer key

1.23.28 Identify Function: GATE CSE 2015 | Set 2 | Question: 11



Consider the following C function.

```
int fun(int n) {
```

```

int x=1, k;
if (n==1) return x;
for (k=1; k<n; ++k)
    x = x + fun(k) * fun (n-k);
return x;
}

```

1

The return value of $fun(5)$ is _____. 2

gatecse-2015-set2 algorithms identify-function recurrence-relation normal numerical-answers 3

Answer key

1.23.29 Identify Function: GATE CSE 2015 | Set 3 | Question: 49



Suppose $c = \langle c[0], \dots, c[k-1] \rangle$ is an array of length k , where all the entries are from the set $\{0, 1\}$. For any positive integers a and n , consider the following pseudocode. 4

```

DOSOMETHING (c, a, n)
z ← 1
for i ← 0 to k-1
    do z ← z2 mod n
    if c[i]=1
        then z ← (z × a) mod n
return z

```

5

If $k = 4$, $c = \langle 1, 0, 1, 1 \rangle$, $a = 2$, and $n = 8$, then the output of $DOSOMETHING(c, a, n)$ is _____. 6

gatecse-2015-set3 algorithms identify-function normal numerical-answers

Answer key

1.23.30 Identify Function: GATE CSE 2019 | Question: 26



Consider the following C function. 7

```

void convert (int n) {
    if (n<0)
        printf("%d", n);
    else {
        convert(n/2);
        printf("%d", n%2);
    }
}

```

8

Which one of the following will happen when the function *convert* is called with any positive integer n as argument? 9

- A. It will print the binary representation of n and terminate
- B. It will print the binary representation of n in the reverse order and terminate
- C. It will print the binary representation of n but will not terminate
- D. It will not print anything and will not terminate

10

gatecse-2019 algorithms identify-function two-marks

Answer key

1.23.31 Identify Function: GATE CSE 2020 | Question: 48



Consider the following C functions.

```
int tob (int b, int* arr) {
    int i;
    for (i = 0; b>0; i++) {
        if (b%2) arr[i] = 1;
        else arr[i] = 0;
        b = b/2;
    }
    return (i);
}
```

```
int pp(int a, int b) {
    int arr[20];
    int i, tot = 1, ex, len;
    ex = a;
    len = tob(b, arr);
    for (i=0; i<len ; i++) {
        if (arr[i] ==1)
            tot = tot * ex;
            ex= ex*ex;
    }
    return (tot) ;
}
```

The value returned by $pp(3, 4)$ is 2.

gatecse-2020 numerical-answers identify-function two-marks

Answer key

1.23.32 Identify Function: GATE CSE 2021 | Set 1 | Question: 48

Consider the following ANSI C function:

```
int SimpleFunction(int Y[], int n, int x)
{
    int total = Y[0], loopIndex;
    for (loopIndex=1; loopIndex<=n-1; loopIndex++)
        total=x*total +Y[loopIndex];
    return total;
}
```

Let Z be an array of 10 elements with $Z[i] = 1$, for all i such that $0 \leq i \leq 9$. The value returned by $\text{SimpleFunction}(Z, 10, 2)$ is _____

gatecse-2021-set1 algorithms numerical-answers identify-function two-marks

Answer key

1.23.33 Identify Function: GATE CSE 2021 | Set 2 | Question: 23

Consider the following ANSI C function:

```
int SomeFunction (int x, int y)
{
    if ((x==1) || (y==1)) return 1;
    if (x==y) return x;
    if (x > y) return SomeFunction(x-y, y);
    if (y > x) return SomeFunction (x, y-x);
}
```

The value returned by $\text{SomeFunction}(15, 255)$ is _____

gatecse-2021-set2 numerical-answers algorithms identify-function output one-mark

Answer key

1.23.34 Identify Function: GATE IT 2005 | Question: 53

The following C function takes two ASCII strings and determines whether one is an anagram of the other. An anagram of a string s is a string obtained by permuting the letters in s .

```
int anagram (char *a, char *b) {
    int count [128], j;
    for (j = 0; j < 128; j++) count[j] = 0;
    j = 0;
    while (a[j] && b[j]) {
        A;
        B;
    }
    for (j = 0; j < 128; j++) if (count[j]) return 0;
    return 1;
}
```


Choose the correct alternative for statements *A* and *B*. ¹

- A. A: count [a[j]]++ and B: count[b[j]]-- ²
- B. A: count [a[j]]++ and B: count[b[j]]++
- C. A: count [a[j++]]++ and B: count[b[j]]--
- D. A: count [a[j]]++ and B: count[b[j++]]--

gateit-2005 normal identify-function

Answer key

1.23.35 Identify Function: GATE IT 2005 | Question: 57



What is the output printed by the following program? ³

```
#include <stdio.h>

int f(int n, int k) {
    if (n == 0) return 0;
    else if (n % 2) return f(n/2, 2*k) + k;
    else return f(n/2, 2*k) - k;
}

int main () {
    printf("%d", f(20, 1));
    return 0;
}
```

- A. 5 B. 8 C. 9 D. 20 ⁵

gateit-2005 algorithms identify-function normal ⁶

Answer key

1.23.36 Identify Function: GATE IT 2006 | Question: 52



The following function computes the value of $\binom{m}{n}$ correctly for all legal values m and n ($m \geq 1, n \geq 0$ and $m > n$). ⁷

```
int func(int m, int n)
{
    if (E) return 1;
    else return(func(m-1, n) + func(m-1, n-1));
}
```

In the above function, which of the following is the correct expression for E? ⁹

- A. $(n == 0) || (m == 1)$ B. $(n == 0) \&\& (m == 1)$ ¹⁰
C. $(n == 0) || (m == n)$ D. $(n == 0) \&\& (m == n)$

gateit-2006 algorithms identify-function normal ¹¹

Answer key

1.23.37 Identify Function: GATE IT 2008 | Question: 82



Consider the code fragment written in C below :

```
void f (int n)
{
    if (n <=1) {
        printf ("%d", n);
    }
    else {
        f (n/2);
        printf ("%d", n%2);
    }
}
```

What does f(173) print?

- A. 010110101 B. 010101101 C. 10110101 D. 10101101

Answer key

1.23.38 Identify Function: GATE IT 2008 | Question: 83



Consider the code fragment written in C below : 1

```
void f (int n)
{
    if (n <= 1) {
        printf ("%d", n);
    }
    else {
        f (n/2);
        printf ("%d", n%2);
    }
}
```

2

Which of the following implementations will produce the same output for $f(173)$ as the above code? 3

P1 4

P2 5

```
void f (int n)
{
    if (n/2) {
        f(n/2);
    }
    printf ("%d", n%2);
}
```

```
void f (int n)
{
    if (n <=1) {
        printf ("%d", n);
    }
    else {
        printf ("%d", n%2);
        f (n/2);
    }
}
```

6

- A. Both $P1$ and $P2$ B. $P2$ only C. $P1$ only D. Neither $P1$ nor $P2$ 7

Answer key

1.24

Insertion Sort (2)

1.24.1 Insertion Sort: GATE CSE 2003 | Question: 22



The usual $\Theta(n^2)$ implementation of Insertion Sort to sort an array uses linear search to identify the position where an element is to be inserted into the already sorted part of the array. If, instead, we use binary search to identify the position, the worst case running time will 9

- A. remain $\Theta(n^2)$ B. become $\Theta(n(\log n)^2)$ 10
C. become $\Theta(n \log n)$ D. become $\Theta(n)$

Answer key 12

1.24.2 Insertion Sort: GATE CSE 2003 | Question: 62



In a permutation $a_1 \dots a_n$, of n distinct integers, an inversion is a pair (a_i, a_j) such that $i < j$ and $a_i > a_j$. What would be the worst case time complexity of the Insertion Sort algorithm, if the inputs are restricted to permutations of $1 \dots n$ with at most n inversions?

- A. $\Theta(n^2)$ B. $\Theta(n \log n)$ C. $\Theta(n^{1.5})$ D. $\Theta(n)$

Answer key

1.25

Inversion (2)

1.25.1 Inversion: GATE CSE 2003 | Question: 61



In a permutation $a_1 \dots a_n$, of n distinct integers, an inversion is a pair (a_i, a_j) such that $i < j$ and $a_i > a_j$. If all permutations are equally likely, what is the expected number of inversions in a randomly chosen permutation of $1 \dots n$?

- A. $\frac{n(n-1)}{2}$ B. $\frac{n(n-1)}{4}$ C. $\frac{n(n+1)}{4}$ D. $2n \lceil \log_2 n \rceil$

gatescse-2003 algorithms sorting inversion normal 3

Answer key

1.25.2 Inversion: GATE DA 2025 | Question: 19



Suppose that insertion sort is applied to the array $[1, 3, 5, 7, 9, 11, x, 15, 13]$ and it takes exactly two swaps to sort the array. Select all possible values of x .

- A. 10 B. 12 C. 14 D. 16

gateda-2025 algorithms insertion-sort sorting multiple-selects easy one-mark inversion 7

Answer key 8

1.26 Linear Probing (2)

Practice Test: Test 1 (5Q) 9

1.26.1 Linear Probing: GATE CSE 2026 | Set 1 | Question: 14



Consider a hash table $P[0, 1, \dots, 10]$ that is initially empty. The hash table is maintained using open addressing with linear probing. The hash function used is $h(x) = (x + 7) \bmod 11$.

Consider the following sequence of insertions performed on P :

1, 13, 22, 15, 11, 24

Which of the following positions in the hash table is/are empty after these insertions are performed?

- A. 0 B. 10 C. 2 D. 1

gatescse-2026-set1 algorithms hashing linear-probing multiple-selects one-mark 15

Answer key

1.26.2 Linear Probing: GATE DA 2025 | Question: 8



Consider a hash table of size 10 with indices $\{0, 1, \dots, 9\}$, with the hash function

$$h(x) = 3x \pmod{10}$$

where linear probing is used to handle collisions. The hash table is initially empty and then the following sequence of keys is inserted into the hash table: 1, 4, 5, 6, 14, 15. The indices where the keys 14 and 15 are stored are, respectively

- A. 2 and 5 B. 2 and 6 C. 4 and 5 D. 4 and 6

gateda-2025 algorithms hashing linear-probing easy one-mark 20

Answer key

1.27 Matrix Chain Ordering (3)

Practice Test: Test 1 (5Q)

1.27.1 Matrix Chain Ordering: GATE CSE 2011 | Question: 38



Four Matrices M_1, M_2, M_3 and M_4 of dimensions $p \times q, q \times r, r \times s$ and $s \times t$ respectively can be multiplied in several ways with different number of total scalar multiplications. For example when multiplied as $((M_1 \times M_2) \times (M_3 \times M_4))$, the total number of scalar multiplications is $pqr + rst + prt$. When multiplied as $((M_1 \times M_2) \times M_3) \times M_4$, the total number of scalar multiplications is $pqr + prs + pst$.

If $p = 10, q = 100, r = 20, s = 5$ and $t = 80$, then the minimum number of scalar multiplications needed is **2**

- A. 248000 B. 44000 C. 19000 D. 25000 **3**

gatecse-2011 algorithms dynamic-programming normal matrix-chain-ordering **4**

Answer key

1.27.2 Matrix Chain Ordering: GATE CSE 2016 | Set 2 | Question: 38



Let A_1, A_2, A_3 and A_4 be four matrices of dimensions $10 \times 5, 5 \times 20, 20 \times 10$ and 10×5 , respectively. The minimum number of scalar multiplications required to find the product $A_1 A_2 A_3 A_4$ using the basic matrix multiplication method is _____.

gatecse-2016-set2 dynamic-programming algorithms matrix-chain-ordering normal numerical-answers **6**

Answer key

1.27.3 Matrix Chain Ordering: GATE CSE 2018 | Question: 31



Assume that multiplying a matrix G_1 of dimension $p \times q$ with another matrix G_2 of dimension $q \times r$ requires pqr scalar multiplications. Computing the product of n matrices $G_1 G_2 G_3 \dots G_n$ can be done by parenthesizing in different ways. Define $G_i G_{i+1}$ as an **explicitly computed pair** for a given paranthesization if they are directly multiplied. For example, in the matrix multiplication chain $G_1 G_2 G_3 G_4 G_5 G_6$ using parenthesization $(G_1 (G_2 G_3)) (G_4 (G_5 G_6))$, $G_2 G_3$ and $G_5 G_6$ are only explicitly computed pairs.

Consider a matrix multiplication chain $F_1 F_2 F_3 F_4 F_5$, where matrices F_1, F_2, F_3, F_4 and F_5 are of dimensions $2 \times 25, 25 \times 3, 3 \times 16, 16 \times 1$ and 1×1000 , respectively. In the parenthesization of $F_1 F_2 F_3 F_4 F_5$ that minimizes the total number of scalar multiplications, the explicitly computed pairs is/are **8**

- A. $F_1 F_2$ and $F_3 F_4$ only B. $F_2 F_3$ only **9**
C. $F_3 F_4$ only D. $F_1 F_2$ and $F_4 F_5$ only

gatecse-2018 algorithms dynamic-programming two-marks matrix-chain-ordering **10**

Answer key

1.28 Maximum Minimum (1)

1.28.1 Maximum Minimum: GATE CSE 2014 | Set 1 | Question: 39



The minimum number of comparisons required to find the minimum and the maximum of 100 numbers is **11**

gatecse-2014-set1 algorithms numerical-answers normal maximum-minimum sorting **12**

Answer key

1.29 Merge Sort (4)

Practice Test: Test 1 (14Q)

1.29.1 Merge Sort: GATE CSE 1999 | Question: 1.14, ISRO2015-42



If one uses straight two-way merge sort algorithm to sort the following elements in ascending order:

20, 47, 15, 8, 9, 4, 40, 30, 12, 17

then the order of these elements after second pass of the algorithm is:

- A. 8, 9, 15, 20, 47, 4, 12, 17, 30, 40
B. 8, 15, 20, 47, 4, 9, 30, 40, 12, 17
C. 15, 20, 47, 4, 8, 9, 12, 30, 40, 17

D. 4, 8, 9, 15, 20, 47, 12, 17, 30, 40¹

gate1999 algorithms merge-sort normal isro2015 sorting²

Answer key

1.29.2 Merge Sort: GATE CSE 2012 | Question: 39

A list of n strings, each of length n , is sorted into lexicographic order using the merge-sort algorithm. The worst case running time of this computation is³

A. $O(n \log n)$ B. $O(n^2 \log n)$ C. $O(n^2 + \log n)$ D. $O(n^2)$ ⁴

gatecse-2012 algorithms sorting normal merge-sort⁵

Answer key

1.29.3 Merge Sort: GATE CSE 2015 | Set 3 | Question: 27

Assume that a mergesort algorithm in the worst case takes 30 seconds for an input of size 64. Which of the following most closely approximates the maximum input size of a problem that can be solved in 6 minutes?⁶

A. 256 B. 512 C. 1024 D. 2018⁷

gatecse-2015-set3 algorithms sorting merge-sort⁸

Answer key

1.29.4 Merge Sort: GATE CSE 2026 | Set 2 | Question: 22

Consider an array $A = [10, 7, 8, 19, 41, 35, 25, 31]$. Suppose the merge sort algorithm is executed on array A to sort it in increasing order. The merge sort algorithm will carry out a total of 7 merge operations.⁹

A merge operation on sorted left array L and sorted right array R is said to be void if the output of the merge operation is the elements of array L followed by the elements of array R .¹⁰

The number of void merge operations among these 7 merge operations is _____. (answer in integer)¹¹

gatecse-2026-set2 algorithms merge-sort numerical-answers one-mark¹²

Answer key¹³

1.30

Merging (2)¹⁴

Practice Test: Test 1 (6Q)

15

1.30.1 Merging: GATE CSE 1995 | Question: 1.16

For merging two sorted lists of sizes m and n into a sorted list of size $m + n$, we require comparisons of¹⁶

A. $O(m)$ B. $O(n)$ C. $O(m + n)$ D. $O(\log m + \log n)$ ¹⁸

gate1995 algorithms sorting normal merging¹⁹

Answer key

1.30.2 Merging: GATE CSE 2014 | Set 2 | Question: 38

Suppose P, Q, R, S, T are sorted sequences having lengths 20, 24, 30, 35, 50 respectively. They are to be merged into a single sequence by merging together two sequences at a time. The number of comparisons that will be needed in the worst case by the optimal algorithm for doing this is _____.

gatecse-2014-set2 algorithms sorting normal numerical-answers merging

Answer key

1.31

Minimum Spanning Tree (35)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (9Q)

1.31.1 Minimum Spanning Tree: GATE CSE 1991 | Question: 03,vi



Kruskal's algorithm for finding a minimum spanning tree of a weighted graph G with n vertices and m edges has the time complexity of:

- A. $O(n^2)$ B. $O(mn)$ C. $O(m+n)$ D. $O(m \log n)$ E. $O(m^2)$

gate1991 algorithms graph-algorithms minimum-spanning-tree time-complexity multiple-selects 3

Answer key

1.31.2 Minimum Spanning Tree: GATE CSE 1992 | Question: 01,ix



Complexity of Kruskal's algorithm for finding the minimum spanning tree of an undirected graph containing n vertices and m edges if the edges are sorted is _____

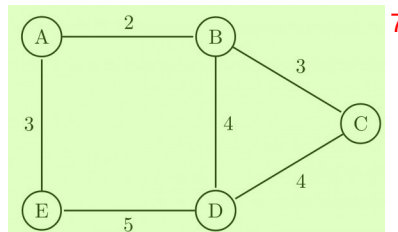
gate1992 minimum-spanning-tree algorithms time-complexity easy fill-in-the-blanks 5

Answer key

1.31.3 Minimum Spanning Tree: GATE CSE 1995 | Question: 22



How many minimum spanning trees does the following graph have? Draw them. (Weights are assigned to edges).



gate1995 algorithms graph-algorithms minimum-spanning-tree easy descriptive 8

Answer key

1.31.4 Minimum Spanning Tree: GATE CSE 1996 | Question: 16



A complete, undirected, weighted graph G is given on the vertex $\{0, 1, \dots, n-1\}$ for any fixed 'n'. Draw the minimum spanning tree of G if

- A. the weight of the edge (u, v) is $|u - v|$ B. the weight of the edge (u, v) is $u + v$

gate1996 algorithms graph-algorithms minimum-spanning-tree normal descriptive 11

Answer key

1.31.5 Minimum Spanning Tree: GATE CSE 1997 | Question: 9



Consider a graph whose vertices are points in the plane with integer co-ordinates (x, y) such that $1 \leq x \leq n$ and $1 \leq y \leq n$, where $n \geq 2$ is an integer. Two vertices (x_1, y_1) and (x_2, y_2) are adjacent iff $|x_1 - x_2| \leq 1$ and $|y_1 - y_2| \leq 1$. The weight of an edge $\{(x_1, y_1), (x_2, y_2)\}$ is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

- A. What is the weight of a minimum weight-spanning tree in this graph? Write only the answer without any explanations. B. What is the weight of a maximum weight-spanning tree in this graph? Write only the answer without any explanations.

gate1997 algorithms minimum-spanning-tree normal descriptive

Answer key

1.31.6 Minimum Spanning Tree: GATE CSE 2000 | Question: 2.18



Let G be an undirected connected graph with distinct edge weights. Let e_{max} be the edge with maximum weight and e_{min} the edge with minimum weight. Which of the following statements is false?

- A. Every minimum spanning tree of G must contain e_{min}
- B. If e_{max} is in a minimum spanning tree, then its removal must disconnect G
- C. No minimum spanning tree contains e_{max}
- D. G has a unique minimum spanning tree

gatecse-2000 algorithms minimum-spanning-tree normal 3

Answer key

1.31.7 Minimum Spanning Tree: GATE CSE 2001 | Question: 15



Consider a weighted undirected graph with vertex set $V = \{n1, n2, n3, n4, n5, n6\}$ and edge set $E = \{(n1, n2, 2), (n1, n3, 8), (n1, n6, 3), (n2, n4, 4), (n2, n5, 12), (n3, n4, 7), (n4, n5, 9), (n4, n6, 4)\}$.

The third value in each tuple represents the weight of the edge specified in the tuple.

- A. List the edges of a minimum spanning tree of the graph.
- B. How many distinct minimum spanning trees does this graph have?
- C. Is the minimum among the edge weights of a minimum spanning tree unique over all possible minimum spanning trees of a graph?
- D. Is the maximum among the edge weights of a minimum spanning tree unique over all possible minimum spanning trees of a graph?

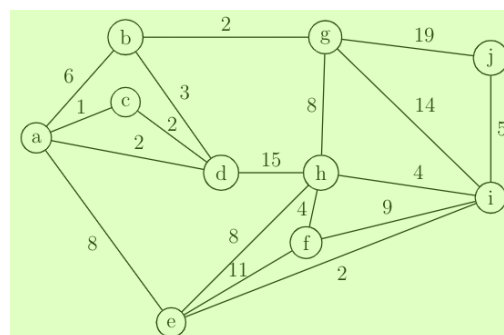
gatecse-2001 algorithms minimum-spanning-tree normal descriptive 8

Answer key

1.31.8 Minimum Spanning Tree: GATE CSE 2003 | Question: 68



What is the weight of a minimum spanning tree of the following graph?



- A. 29
- B. 31
- C. 38
- D. 41

gatecse-2003 algorithms minimum-spanning-tree normal 12

Answer key

1.31.9 Minimum Spanning Tree: GATE CSE 2005 | Question: 6



An undirected graph G has n nodes. its adjacency matrix is given by an $n \times n$ square matrix whose (i) diagonal elements are 0's and (ii) non-diagonal elements are 1's. Which one of the following is TRUE?

- A. Graph G has no minimum spanning tree (MST)
- B. Graph G has unique MST of cost $n - 1$
- C. Graph G has multiple distinct MSTs, each of cost $n - 1$
- D. Graph G has multiple spanning trees of different costs

Answer key

1.31.10 Minimum Spanning Tree: GATE CSE 2006 | Question: 11



Consider a weighted complete graph G on the vertex set $\{v_1, v_2, \dots, v_n\}$ such that the weight of the edge (v_i, v_j) is $2|i - j|$. The weight of a minimum spanning tree of G is:

- A. $n - 1$ B. $2n - 2$ C. $\binom{n}{2}$ D. n^2

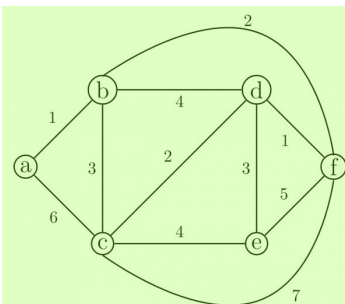
gatecse-2006 algorithms minimum-spanning-tree normal 3

Answer key

1.31.11 Minimum Spanning Tree: GATE CSE 2006 | Question: 47



Consider the following graph:



Which one of the following cannot be the sequence of edges added, in that order, to a minimum spanning tree using Kruskal's algorithm?

- A. $(a - b), (d - f), (b - f), (d - c), (d - e)$ B. $(a - b), (d - f), (d - c), (b - f), (d - e)$
 C. $(d - f), (a - b), (d - c), (b - f), (d - e)$ D. $(d - f), (a - b), (b - f), (d - e), (d - c)$

gatecse-2006 algorithms graph-algorithms minimum-spanning-tree normal 8

Answer key

1.31.12 Minimum Spanning Tree: GATE CSE 2007 | Question: 49



Let w be the minimum weight among all edge weights in an undirected connected graph. Let e be a specific edge of weight w . Which of the following is FALSE?

- A. There is a minimum spanning tree containing e
 B. If e is not in a minimum spanning tree T , then in the cycle formed by adding e to T , all edges have the same weight.
 C. Every minimum spanning tree has an edge of weight w
 D. e is present in every minimum spanning tree

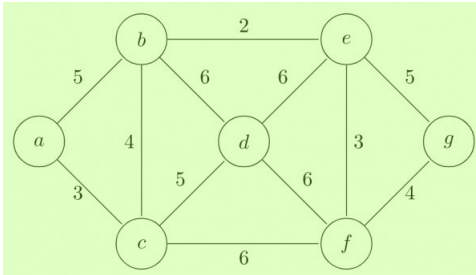
gatecse-2007 algorithms minimum-spanning-tree normal 11

Answer key

1.31.13 Minimum Spanning Tree: GATE CSE 2009 | Question: 38



Consider the following graph:



1

Which one of the following is NOT the sequence of edges added to the minimum spanning tree using Kruskal's algorithm? ²

- A. (b, e) (e, f) (a, c) (b, c) (f, g) (c, d) ³
- B. (b, e) (e, f) (a, c) (f, g) (b, c) (c, d)
- C. (b, e) (a, c) (e, f) (b, c) (f, g) (c, d)
- D. (b, e) (e, f) (b, c) (a, c) (f, g) (c, d)

gatecse-2009 algorithms minimum-spanning-tree normal

Answer key ⁴

1.31.14 Minimum Spanning Tree: GATE CSE 2010 | Question: 50



Consider a complete undirected graph with vertex set $\{0, 1, 2, 3, 4\}$. Entry W_{ij} in the matrix W below is the weight of the edge $\{i, j\}$ ⁵

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix} \quad 6$$

What is the minimum possible weight of a spanning tree T in this graph such that vertex 0 is a leaf node in the tree T ? ⁷

- A. 7
- B. 8
- C. 9
- D. 10 ⁸

gatecse-2010 algorithms minimum-spanning-tree normal ⁹

Answer key ¹⁰

1.31.15 Minimum Spanning Tree: GATE CSE 2010 | Question: 51



Consider a complete undirected graph with vertex set $\{0, 1, 2, 3, 4\}$. Entry W_{ij} in the matrix W below is the weight of the edge $\{i, j\}$ ¹¹

$$W = \begin{pmatrix} 0 & 1 & 8 & 1 & 4 \\ 1 & 0 & 12 & 4 & 9 \\ 8 & 12 & 0 & 7 & 3 \\ 1 & 4 & 7 & 0 & 2 \\ 4 & 9 & 3 & 2 & 0 \end{pmatrix} \quad 12$$

What is the minimum possible weight of a path P from vertex 1 to vertex 2 in this graph such that P contains at most 3 edges? ¹³

- A. 7
- B. 8
- C. 9
- D. 10 ¹⁴

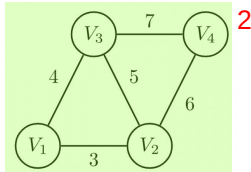
gatecse-2010 normal algorithms minimum-spanning-tree

Answer key

1.31.16 Minimum Spanning Tree: GATE CSE 2011 | Question: 54



An undirected graph $G(V, E)$ contains n ($n > 2$) nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 < |i - j| \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. A sample graph with $n = 4$ is shown below.



What will be the cost of the minimum spanning tree (MST) of such a graph with n nodes? **3**

- A. $\frac{1}{12}(11n^2 - 5n)$ B. $n^2 - n + 1$ C. $6n - 11$ D. $2n + 1$ **4**

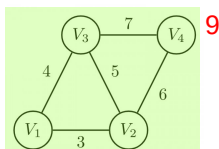
gatecse-2011 algorithms graph-algorithms minimum-spanning-tree normal **5**

Answer key **6**

1.31.17 Minimum Spanning Tree: GATE CSE 2011 | Question: 55



An undirected graph $G(V, E)$ contains n ($n > 2$) nodes named $v_1, v_2, \dots, v_n$. Two nodes v_i, v_j are connected if and only if $0 < |i - j| \leq 2$. Each edge (v_i, v_j) is assigned a weight $i + j$. A sample graph with $n = 4$ is shown below.



The length of the path from v_5 to v_6 in the MST of previous question with $n = 10$ is **10**

- A. 11 B. 25 C. 31 D. 41 **11**

gatecse-2011 algorithms graph-algorithms minimum-spanning-tree normal **12**

Answer key **13**

1.31.18 Minimum Spanning Tree: GATE CSE 2012 | Question: 29



Let G be a weighted graph with edge weights greater than one and G' be the graph constructed by squaring the weights of edges in G . Let T and T' be the minimum spanning trees of G and G' , respectively, with total weights t and t' . Which of the following statements is **TRUE**? **14**

- A. $T' = T$ with total weight $t' = t^2$ B. $T' = T$ with total weight $t' < t^2$ **15**
C. $T' \neq T$ but total weight $t' = t^2$ D. None of the above

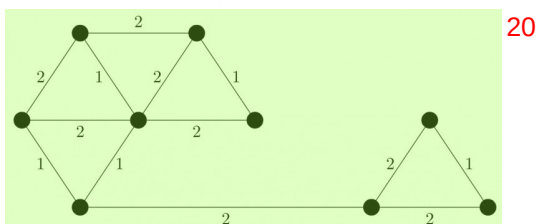
gatecse-2012 algorithms minimum-spanning-tree normal marks-to-all **16**

Answer key **17**

1.31.19 Minimum Spanning Tree: GATE CSE 2014 | Set 2 | Question: 52



The number of distinct minimum spanning trees for the weighted graph below is **19**



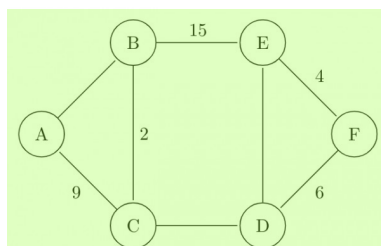
gatecse-2014-set2 algorithms minimum-spanning-tree numerical-answers normal

Answer key

1.31.20 Minimum Spanning Tree: GATE CSE 2015 | Set 1 | Question: 43



The graph shown below has 8 edges with distinct integer edge weights. The minimum spanning tree (**MST**) is of weight 36 and contains the edges: $\{(A, C), (B, C), (B, E), (E, F), (D, F)\}$. The edge weights of only those edges which are in the **MST** are given in the figure shown below. The minimum possible sum of weights of all 8 edges of this graph is _____.



gatecse-2015-set1 algorithms minimum-spanning-tree normal numerical-answers

Answer key

1.31.21 Minimum Spanning Tree: GATE CSE 2015 | Set 3 | Question: 40



Let G be a connected undirected graph of 100 vertices and 300 edges. The weight of a minimum spanning tree of G is 500. When the weight of each edge of G is increased by five, the weight of a minimum spanning tree becomes _____.

gatecse-2015-set3 algorithms minimum-spanning-tree easy numerical-answers

Answer key

1.31.22 Minimum Spanning Tree: GATE CSE 2016 | Set 1 | Question: 14



Let G be a weighted connected undirected graph with distinct positive edge weights. If every edge weight is increased by the same value, then which of the following statements is/are TRUE?

- P : Minimum spanning tree of G does not change.
- Q : Shortest path between any pair of vertices does not change.

A. P only B. Q only C. Neither P nor Q D. Both P and Q

gatecse-2016-set1 algorithms minimum-spanning-tree normal

Answer key

1.31.23 Minimum Spanning Tree: GATE CSE 2016 | Set 1 | Question: 39



Let G be a complete undirected graph on 4 vertices, having 6 edges with weights being 1, 2, 3, 4, 5, and 6. The maximum possible weight that a minimum weight spanning tree of G can have is _____.

gatecse-2016-set1 algorithms minimum-spanning-tree normal numerical-answers

Answer key

1.31.24 Minimum Spanning Tree: GATE CSE 2016 | Set 1 | Question: 40



$G = (V, E)$ is an undirected simple graph in which each edge has a distinct weight, and e is a particular edge of G . Which of the following statements about the minimum spanning trees (**MSTs**) of G is/are TRUE?

- I. If e is the lightest edge of some cycle in G , then every MST of G includes e .
- II. If e is the heaviest edge of some cycle in G , then every MST of G excludes e .

A. I only. B. II only. C. Both I and II. D. Neither I nor II.

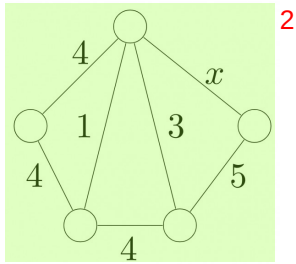
gatecse-2016-set1 algorithms minimum-spanning-tree normal

Answer key

1.31.25 Minimum Spanning Tree: GATE CSE 2018 | Question: 47



Consider the following undirected graph G :¹



Choose a value for x that will maximize the number of minimum weight spanning trees (MWSTs) of G . The number³ of MWSTs of G for this value of x is ____.

gatecse-2018 algorithms graph-algorithms minimum-spanning-tree numerical-answers two-marks⁴

Answer key

1.31.26 Minimum Spanning Tree: GATE CSE 2020 | Question: 31



Let $G = (V, E)$ be a weighted undirected graph and let T be a Minimum Spanning Tree (MST) of G maintained using adjacency lists. Suppose a new weighed edge $(u, v) \in V \times V$ is added to G . The worst case time complexity of determining if T is still an MST of the resultant graph is⁵

- A. $\Theta(|E| + |V|)$ B. $\Theta(|E| |V|)$ ⁶
C. $\Theta(E \log V)$ D. $\Theta(V)$

gatecse-2020 algorithms minimum-spanning-tree graph-algorithms two-marks⁷

Answer key

1.31.27 Minimum Spanning Tree: GATE CSE 2020 | Question: 49



Consider a graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_{100}\}$, $E = \{(v_i, v_j) \mid 1 \leq i < j \leq 100\}$, and⁸ weight of the edge (v_i, v_j) is $|i - j|$. The weight of minimum spanning tree of G is ____.

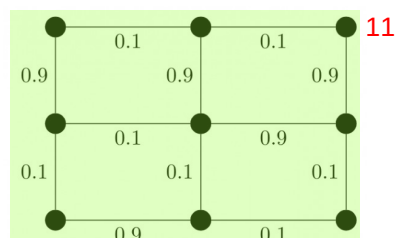
gatecse-2020 numerical-answers algorithms graph-algorithms two-marks minimum-spanning-tree⁹

Answer key

1.31.28 Minimum Spanning Tree: GATE CSE 2021 | Set 1 | Question: 17



Consider the following undirected graph with edge weights as shown:¹⁰



The number of minimum-weight spanning trees of the graph is ____.¹²

gatecse-2021-set1 algorithms graph-algorithms minimum-spanning-tree numerical-answers one-mark¹³

Answer key

1.31.29 Minimum Spanning Tree: GATE CSE 2021 | Set 2 | Question: 1



Let G be a connected undirected weighted graph. Consider the following two statements.

- S_1 : There exists a minimum weight edge in G which is present in every minimum spanning tree of G .
- S_2 : If every edge in G has distinct weight, then G has a unique minimum spanning tree.

Which one of the following options is correct? 1

- A. Both S_1 and S_2 are true
C. S_1 is false and S_2 is true

- B. S_1 is true and S_2 is false 2
D. Both S_1 and S_2 are false

gatecse-2021-set2 algorithms graph-algorithms minimum-spanning-tree one-mark 3

Answer key

1.31.30 Minimum Spanning Tree: GATE CSE 2022 | Question: 39



Consider a simple undirected weighted graph G , all of whose edge weights are distinct. Which of the following statements about the minimum spanning trees of G is/are TRUE? 4

- A. The edge with the second smallest weight is always part of any minimum spanning tree of G .
B. One or both of the edges with the third smallest and the fourth smallest weights are part of any minimum spanning tree of G .
C. Suppose $S \subseteq V$ be such that $S \neq \phi$ and $S \neq V$. Consider the edge with the minimum weight such that one of its vertices is in S and the other in $V \setminus S$. Such an edge will always be part of any minimum spanning tree of G .
D. G can have multiple minimum spanning trees. 5

gatecse-2022 algorithms minimum-spanning-tree multiple-selects two-marks 6

Answer key

1.31.31 Minimum Spanning Tree: GATE CSE 2022 | Question: 48



Let $G(V, E)$ be a directed graph, where $V = \{1, 2, 3, 4, 5\}$ is the set of vertices and E is the set of directed edges, as defined by the following adjacency matrix A . 7

$$A[i][j] = \begin{cases} 1, & 1 \leq j \leq i \leq 5 \\ 0, & \text{otherwise} \end{cases} 9$$

$A[i][j] = 1$ indicates a directed edge from node i to node j . A *directed spanning tree* of G , rooted at $r \in V$, is defined as a subgraph T of G such that the undirected version of T is a tree, and T contains a directed path from r to every other vertex in V . The number of such directed spanning trees rooted at vertex 5 is 10

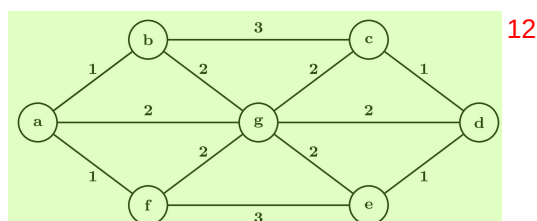
gatecse-2022 numerical-answers algorithms minimum-spanning-tree two-marks

Answer key

1.31.32 Minimum Spanning Tree: GATE CSE 2024 | Set 2 | Question: 49



The number of distinct minimum-weight spanning trees of the following graph is 11



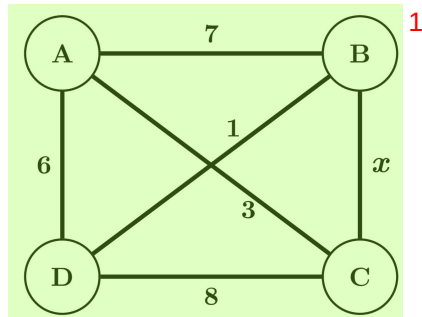
gatecse-2024-set2 numerical-answers algorithms minimum-spanning-tree two-marks

Answer key

1.31.33 Minimum Spanning Tree: GATE CSE 2025 | Set 1 | Question: 54



The maximum value of x such that the edge between the nodes B and C is included in every minimum spanning tree of the given graph is _____. (answer in integer)



gatecse2025-set1 algorithms minimum-spanning-tree numerical-answers easy two-marks

Answer key

1.31.34 Minimum Spanning Tree: GATE CSE 2026 | Set 1 | Question: 39



Let $G(V, E)$ be a simple, undirected, edge-weighted graph with unique edge weights. Which of the following statements about the minimum spanning trees (MST) of G is/are true?

- A. In every cycle C of G , the edge with the largest weight in C is not in any MST
- B. In every cycle C of G , the edge with the smallest weight in C is in every MST
- C. For every vertex $v \in V$, the edge with the largest weight incident on v is not in any MST
- D. For every vertex $v \in V$, the edge with the smallest weight incident on v is in every MST

gatecse-2026-set1 two-marks algorithms minimum-spanning-tree multiple-selects

Answer key

1.31.35 Minimum Spanning Tree: GATE IT 2005 | Question: 52



Let G be a weighted undirected graph and e be an edge with maximum weight in G . Suppose there is a minimum weight spanning tree in G containing the edge e . Which of the following statements is always TRUE?

- A. There exists a cutset in G having all edges of maximum weight.
- B. There exists a cycle in G having all edges of maximum weight.
- C. Edge e cannot be contained in a cycle.
- D. All edges in G have the same weight.

gateit-2005 algorithms minimum-spanning-tree normal

Answer key

1.32 Number of Swap (1)

1.32.1 Number of Swap: GATE CSE 2025 | Set 1 | Question: 23



The pseudocode of a function `fun ()` is given below:

```
fun(int A[0,...,n-1]) {
  for i=0 to n-2
    for j=0 to n-i-2
      if (A[j]>A[j+1])
        then swap A[j] and A[j+1]
}
```

Let $A[0, \dots, 29]$ be an array storing 30 distinct integers in descending order. The number of swap operations that will be performed, if the function `fun ()` is called with $A[0, \dots, 29]$ as argument, is _____. (Answer in integer)

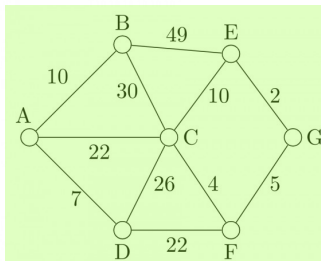
gatecse2025-set1 data-structures array number-of-swap numerical-answers one-mark

Answer key

1.33.1 Prims Algorithm: GATE IT 2004 | Question: 56



Consider the undirected graph below: 1



Using Prim's algorithm to construct a minimum spanning tree starting with node A, which one of the following sequences of edges represents a possible order in which the edges would be added to construct the minimum spanning tree? 3

- A. (E, G), (C, F), (F, G), (A, D), (A, B), (A, C) 4
 B. (A, D), (A, B), (A, C), (C, F), (G, E), (F, G)
 C. (A, B), (A, D), (D, F), (F, G), (G, E), (F, C)
 D. (A, D), (A, B), (D, F), (F, C), (F, G), (G, E)

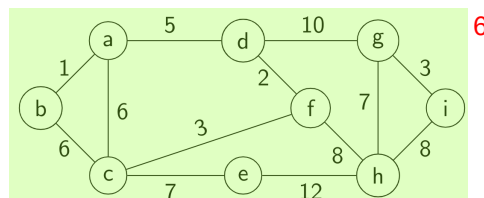
gateit-2004 algorithms graph-algorithms normal prims-algorithm

Answer key

1.33.2 Prims Algorithm: GATE IT 2008 | Question: 45



For the undirected, weighted graph given below, which of the following sequences of edges represents a correct execution of Prim's algorithm to construct a Minimum Spanning Tree? 5



- A. (a, b), (d, f), (f, c), (g, i), (d, a), (g, h), (c, e), (f, h) 7
 B. (c, e), (c, f), (f, d), (d, a), (a, b), (g, h), (h, f), (g, i)
 C. (d, f), (f, c), (d, a), (a, b), (c, e), (f, h), (g, h), (g, i)
 D. (h, g), (g, i), (h, f), (f, c), (f, d), (d, a), (a, b), (c, e)

gateit-2008 algorithms graph-algorithms minimum-spanning-tree normal prims-algorithm

Answer key

Practice Tests: Test 1 (15Q) Test 2 (2Q)

1.34.1 Quick Sort: GATE CSE 1987 | Question: 1-xviii



Let P be a quicksort program to sort numbers in ascending order. Let t_1 and t_2 be the time taken by the program for the inputs $[1\ 2\ 3\ 4]$ and $[5\ 4\ 3\ 2\ 1]$, respectively. Which of the following holds?

- A. $t_1 = t_2$ B. $t_1 > t_2$
 C. $t_1 < t_2$ D. $t_1 = t_2 + 5 \log 5$

gate1987 algorithms sorting quick-sort

Answer key

1.34.2 Quick Sort: GATE CSE 1989 | Question: 9



An input file has 10 records with keys as given below: ¹

25 7 34 2 70 9 61 16 49 19

This is to be sorted in non-decreasing order.

- Sort the input file using QUICKSORT by correctly positioning the first element of the file/subfile. Show the subfiles obtained at all intermediate steps. Use square brackets to demarcate subfiles. ²
- Sort the input file using 2-way- MERGESORT showing all major intermediate steps. Use square brackets to demarcate subfiles.

gate1989 descriptive algorithms sorting quick-sort

Answer key

1.34.3 Quick Sort: GATE CSE 1992 | Question: 03,iv



Assume that the last element of the set is used as partition element in Quicksort. If n distinct elements from the set $[1 \dots n]$ are to be sorted, give an input for which Quicksort takes maximum time. ³

gate1992 algorithms sorting easy quick-sort descriptive ⁴

Answer key

1.34.4 Quick Sort: GATE CSE 1996 | Question: 2.15



Quick-sort is run on two inputs shown below to sort in ascending order taking first element as pivot ⁵

- $1, 2, 3, \dots, n$ ⁶
- $n, n-1, n-2, \dots, 2, 1$

Let C_1 and C_2 be the number of comparisons made for the inputs (i) and (ii) respectively. Then, ⁷

- | | |
|----------------|---|
| A. $C_1 < C_2$ | B. $C_1 > C_2$ ⁸ |
| C. $C_1 = C_2$ | D. we cannot say anything for arbitrary n |

gate1996 algorithms sorting normal quick-sort ⁹

Answer key

1.34.5 Quick Sort: GATE CSE 2001 | Question: 1.14



Randomized quicksort is an extension of quicksort where the pivot is chosen randomly. What is the worst case complexity of sorting n numbers using Randomized quicksort? ¹⁰

- | | | | |
|-----------|------------------|-------------|--------------------------|
| A. $O(n)$ | B. $O(n \log n)$ | C. $O(n^2)$ | D. $O(n!)$ ¹¹ |
|-----------|------------------|-------------|--------------------------|

gatecse-2001 algorithms sorting time-complexity easy quick-sort ¹²

Answer key

1.34.6 Quick Sort: GATE CSE 2006 | Question: 52



The median of n elements can be found in $O(n)$ time. Which one of the following is correct about the complexity of quick sort, in which median is selected as pivot? ¹³

- | | |
|------------------|-------------------------------------|
| A. $\Theta(n)$ | B. $\Theta(n \log n)$ ¹⁵ |
| C. $\Theta(n^2)$ | D. $\Theta(n^3)$ |

gatecse-2006 algorithms sorting easy quick-sort ¹⁶

Answer key

1.34.7 Quick Sort: GATE CSE 2008 | Question: 43



Consider the Quicksort algorithm. Suppose there is a procedure for finding a pivot element which splits the list into two sub-lists each of which contains at least one-fifth of the elements. Let $T(n)$ be the number of comparisons required to sort n elements. Then ¹⁷

- A. $T(n) \leq 2T(n/5) + n$ B. $T(n) \leq T(n/5) + T(4n/5) + n$ 1
C. $T(n) \leq 2T(4n/5) + n$ D. $T(n) \leq 2T(n/2) + n$

gatese-2008 algorithms sorting easy quick-sort 2

Answer key 3

1.34.8 Quick Sort: GATE CSE 2009 | Question: 39

In quick-sort, for sorting n elements, the $(n/4)^{th}$ smallest element is selected as pivot using an $O(n)$ time algorithm. What is the worst case time complexity of the quick sort?

- A. $\Theta(n)$ B. $\Theta(n \log n)$ 6
C. $\Theta(n^2)$ D. $\Theta(n^2 \log n)$

gatese-2009 algorithms sorting normal quick-sort 7

Answer key 8

1.34.9 Quick Sort: GATE CSE 2014 | Set 1 | Question: 14

Let P be quicksort program to sort numbers in ascending order using the first element as the pivot. Let t_1 and t_2 be the number of comparisons made by P for the inputs $[1\ 2\ 3\ 4\ 5]$ and $[4\ 1\ 5\ 3\ 2]$ respectively. Which one of the following holds?

- A. $t_1 = 5$ B. $t_1 < t_2$ C. $t_1 > t_2$ D. $t_1 = t_2$ 11

gatese-2014-set1 algorithms sorting easy quick-sort 12

Answer key

1.34.10 Quick Sort: GATE CSE 2014 | Set 3 | Question: 14

You have an array of n elements. Suppose you implement quicksort by always choosing the central element of the array as the pivot. Then the tightest upper bound for the worst case performance is

- A. $O(n^2)$ B. $O(n \log n)$ C. $\Theta(n \log n)$ D. $O(n^3)$ 15

gatese-2014-set3 algorithms sorting easy quick-sort 16

Answer key

1.34.11 Quick Sort: GATE CSE 2015 | Set 1 | Question: 2

Which one of the following is the recurrence equation for the worst case time complexity of the quick sort algorithm for sorting $n (\geq 2)$ numbers? In the recurrence equations given in the options below, c is a constant.

- A. $T(n) = 2T(n/2) + cn$ B. $T(n) = T(n-1) + T(1) + cn$ 19
C. $T(n) = 2T(n-1) + cn$ D. $T(n) = T(n/2) + cn$

gatese-2015-set1 algorithms recurrence-relation sorting easy quick-sort 20

Answer key

1.34.12 Quick Sort: GATE CSE 2015 | Set 2 | Question: 45

Suppose you are provided with the following function declaration in the C programming language. 22

```
int partition(int a[], int n); 23
```

The function treats the first element of $a[]$ as a pivot and rearranges the array so that all elements less than or equal to the pivot is in the left part of the array, and all elements greater than the pivot is in the right part. In addition, it moves the pivot so that the pivot is the last element of the left part. The return value is the number of elements in the left part. 24

The following partially given function in the C programming language is used to find the k^{th} smallest element in an array $a[]$ of size n using the partition function. We assume $k \leq n$. 25

```
int kth_smallest(int a[], int n, int k) 26
{
    int left_end = partition(a, n);
```

```

if (left_end+1==k) {
    return a[left_end];
}
if (left_end+1 > k) {
    return kth_smallest (_____);
} else {
    return kth_smallest (_____);
}
}

```

1

The missing arguments lists are respectively 2

- A. $(a, \text{left_end}, k)$ and $(a + \text{left_end} + 1, n - \text{left_end} - 1, k - \text{left_end} - 1)$
- B. $(a, \text{left_end}, k)$ and $(a, n - \text{left_end} - 1, k - \text{left_end} - 1)$
- C. $(a + \text{left_end} + 1, n - \text{left_end} - 1, k - \text{left_end} - 1)$ and $(a, \text{left_end}, k)$
- D. $(a, n - \text{left_end} - 1, k - \text{left_end} - 1)$ and $(a, \text{left_end}, k)$

3

gatecse-2015-set2 algorithms normal sorting quick-sort

Answer key

1.34.13 Quick Sort: GATE CSE 2019 | Question: 20

An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____

gatecse-2019 numerical-answers algorithms quick-sort probability one-mark

5

Answer key

1.34.14 Quick Sort: GATE DA 2026 | Question: 5

Consider that the quick sort algorithm is used to sort an array of n distinct randomly ordered elements. In every call, the pivot is chosen as the first element of the current subarray.

Let $T(n)$ denote the expected time to sort the array. Assume that the time to partition is linear in the size of the current subarray.

Which of the following recurrence relations correctly represents $T(n)$ in this scenario? 8

- A. $T(n) = T(1) + T(n - 1) + O(n)$
- B. $T(n) = T\left(\frac{n}{4}\right) + T\left(\frac{3n}{4}\right) + O(n)$
- C. $T(n) = 2T\left(\frac{n}{2}\right) + O(n)$
- D. $T(n) = \frac{1}{n} \sum_{k=0}^{n-1} [T(k) + T(n - k - 1)] + O(n)$

9

gateda-2026 algorithms quick-sort recurrence-relation one-mark

10

Answer key

1.34.15 Quick Sort: GATE DS&AI 2024 | Question: 20

Consider sorting the following array of integers in ascending order using an inplace Quicksort algorithm that uses the last element as the pivot.

60	70	80	90	100
----	----	----	----	-----

12

The minimum number of swaps performed during this Quicksort is _____ 13

gate-ds-ai-2024 numerical-answers algorithms quick-sort one-mark

14

Answer key

1.35

Recurrence Relation (36)

Practice Tests: Test 1 (15Q) Test 2 (14Q)

1.35.1 Recurrence Relation: GATE CSE 1987 | Question: 10a



Solve the recurrence equations: **1**

- $T(n) = T(n-1) + n$ **2**
- $T(1) = 1$

gate1987 algorithms recurrence-relation descriptive **3**

Answer key **4**

1.35.2 Recurrence Relation: GATE CSE 1988 | Question: 13iv



Solve the recurrence equations: **5**

- $T(n) = T\left(\frac{n}{2}\right) + 1$ **6**
- $T(1) = 1$

gate1988 descriptive algorithms recurrence-relation **7**

Answer key **8**

1.35.3 Recurrence Relation: GATE CSE 1989 | Question: 13b



Find a solution to the following recurrence equation: **9**

- $T(n) = \sqrt{n} + T\left(\frac{n}{2}\right)$
- $T(1) = 1$

gate1989 descriptive algorithms recurrence-relation **10**

Answer key **11**

1.35.4 Recurrence Relation: GATE CSE 1990 | Question: 17a



Express $T(n)$ in terms of the harmonic number $H_n = \sum_{i=1}^n \frac{1}{i}$, $n \geq 1$, where $T(n)$ satisfies the recurrence relation,

$$T(n) = \frac{n+1}{n}T(n-1) + 1, \text{ for } n \geq 2 \text{ and } T(1) = 1 \quad \mathbf{13}$$

What is the asymptotic behaviour of $T(n)$ as a function of n ? **14**

gate1990 descriptive algorithms recurrence-relation **15**

Answer key

1.35.5 Recurrence Relation: GATE CSE 1992 | Question: 07a



Consider the function $F(n)$ for which the pseudocode is given below : **16**

```
Function F(n)
begin
F1 ← 1
if (n=1) then F ← 3
else
  For i = 1 to n do
  begin
    C ← 0
    For j = 1 to n-1 do
    begin C ← C + 1 end
    F1 = F1 * C
  end
F = F1
end
```

17

[n is a positive integer greater than zero] 1

A. Derive a recurrence relation for $F(n)$. 2

gate1992 algorithms recurrence-relation descriptive

Answer key

1.35.6 Recurrence Relation: GATE CSE 1992 | Question: 07b



Consider the function $F(n)$ for which the pseudocode is given below : 3

```
Function F(n)
begin
F1 ← 1
if(n=1) then F ← 3
else
  For i = 1 to n do
    begin
      C ← 0
      For j = 1 to n - 1 do
        begin C ← C + 1 end
      F1 = F1 * C
    end
  end
F = F1
end
```

4

[n is a positive integer greater than zero] 5

B. Solve the recurrence relation for a closed form solution of $F(n)$. 6

gate1992 algorithms recurrence-relation descriptive

Answer key

1.35.7 Recurrence Relation: GATE CSE 1993 | Question: 15



Consider the recursive algorithm given below: 7

```
procedure bubblesort (n); 8
var i,j: index; temp : item;
begin
  for i:=1 to n-1 do
    if A[i] > A[i+1] then
      begin
        temp := A[i];
        A[i] := A[i+1];
        A[i+1] := temp;
      end;
    bubblesort (n-1)
  end
```

Let a_n be the number of times the 'if...then...' statement gets executed when the algorithm is run with value n . Set up the recurrence relation by defining a_n in terms of a_{n-1} . Solve for a_n . 9

gate1993 algorithms recurrence-relation normal descriptive 10

Answer key

1.35.8 Recurrence Relation: GATE CSE 1994 | Question: 1.7, ISRO2017-14



The recurrence relation that arises in relation with the complexity of binary search is:

- | | |
|--|---|
| A. $T(n) = 2T\left(\frac{n}{2}\right) + k$, k is a constant | B. $T(n) = T\left(\frac{n}{2}\right) + k$, k is a constant |
| C. $T(n) = T\left(\frac{n}{2}\right) + \log n$ | D. $T(n) = T\left(\frac{n}{2}\right) + n$ |

gate1994 algorithms recurrence-relation easy isro2017

Answer key

1.35.9 Recurrence Relation: GATE CSE 1996 | Question: 2.12



The recurrence relation ¹

- $T(1) = 2$ ²
- $T(n) = 3T(\frac{n}{4}) + n$

has the solution $T(n)$ equal to ³

- A. $O(n)$ B. $O(\log n)$ C. $O(n^{\frac{3}{4}})$ D. None of the above ⁴

gate1996 algorithms recurrence-relation normal ⁵

Answer key ⁶

1.35.10 Recurrence Relation: GATE CSE 1997 | Question: 15



Consider the following function. ⁷

```
Function F(n, m:integer):integer;
begin
  if (n<=0) or (m<=0) then F:=1
  else
    F:=F(n-1, m) + F(n-1, m-1);
end;
```

⁸

Use the recurrence relation $\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$ to answer the following questions. Assume that n, m are positive integers. Write only the answers without any explanation. ⁹

- a. What is the value of $F(n, 2)$? ¹⁰
 b. What is the value of $F(n, m)$?
 c. How many recursive calls are made to the function F , including the original call, when evaluating $F(n, m)$.

gate1997 algorithms recurrence-relation descriptive

Answer key ¹¹

1.35.11 Recurrence Relation: GATE CSE 1997 | Question: 4.6



Let $T(n)$ be the function defined by $T(1) = 1$, $T(n) = 2T(\lfloor \frac{n}{2} \rfloor) + \sqrt{n}$ for $n \geq 2$. ¹²
 Which of the following statements is true?

- A. $T(n) = O(\sqrt{n})$ B. $T(n) = O(n)$ ¹³
 C. $T(n) = O(\log n)$ D. None of the above

gate1997 algorithms recurrence-relation normal ¹⁴

Answer key ¹⁵

1.35.12 Recurrence Relation: GATE CSE 1998 | Question: 6a



Solve the following recurrence relation

$$x_n = 2x_{n-1} - 1, n > 1$$

$$x_1 = 2$$

gate1998 algorithms recurrence-relation descriptive

Answer key

1.35.13 Recurrence Relation: GATE CSE 2002 | Question: 1.3



The solution to the recurrence equation $T(2^k) = 3T(2^{k-1}) + 1, T(1) = 1$ is

- A. 2^k B. $\frac{(3^{k+1}-1)}{2}$ C. $3^{\log_2 k}$ D. $2^{\log_3 k}$

Answer key

1.35.14 Recurrence Relation: GATE CSE 2002 | Question: 2.11

The running time of the following algorithm ¹

Procedure $A(n)$

If $n \leq 2$ return (1) else return ($A(\lceil \sqrt{n} \rceil)$);

is best described by

- A. $O(n)$ B. $O(\log n)$ C. $O(\log \log n)$ D. $O(1)$ ²

Answer key

1.35.15 Recurrence Relation: GATE CSE 2003 | Question: 35

Consider the following recurrence relation ⁴

$$T(1) = 1$$

$$T(n+1) = T(n) + \lfloor \sqrt{n+1} \rfloor \text{ for all } n \geq 1$$

The value of $T(m^2)$ for $m \geq 1$ is ⁵

- A. $\frac{m}{6}(21m - 39) + 4$ B. $\frac{m}{6}(4m^2 - 3m + 5)$ ⁶
 C. $\frac{m}{2}(3m^{2.5} - 11m + 20) - 5$ D. $\frac{m}{6}(5m^3 - 34m^2 + 137m - 104) + \frac{5}{6}$

Answer key

1.35.16 Recurrence Relation: GATE CSE 2004 | Question: 83, ISRO2015-40

The time complexity of the following C function is (assume $n > 0$) ⁸

```
int recursive (int n) {
    if(n == 1)
        return (1);
    else
        return (recursive (n-1) + recursive (n-1));
}
```

- A. $O(n)$ B. $O(n \log n)$ C. $O(n^2)$ D. $O(2^n)$ ¹⁰

Answer key ¹¹

1.35.17 Recurrence Relation: GATE CSE 2004 | Question: 84

The recurrence equation ¹²

$$T(1) = 1$$

$$T(n) = 2T(n-1) + n, n \geq 2$$

evaluates to ¹³

- A. $2^{n+1} - n - 2$ B. $2^n - n$ C. $2^{n+1} - 2n - 2$ D. $2^n + n$ ¹⁴

Answer key ¹⁶

1.35.18 Recurrence Relation: GATE CSE 2006 | Question: 51, ISRO2016-34

Consider the following recurrence:

$$T(n) = 2T(\sqrt{n}) + 1, T(1) = 1$$

Which one of the following is true? ¹

- A. $T(n) = \Theta(\log \log n)$ B. $T(n) = \Theta(\log n)$ ²
C. $T(n) = \Theta(\sqrt{n})$ D. $T(n) = \Theta(n)$

algorithms recurrence-relation isro2016 gatecse-2006 ³

Answer key 

1.35.19 Recurrence Relation: GATE CSE 2008 | Question: 78



Let x_n denote the number of binary strings of length n that contain no consecutive 0s. ⁴

Which of the following recurrences does x_n satisfy?

- A. $x_n = 2x_{n-1}$ B. $x_n = x_{\lfloor n/2 \rfloor} + 1$ ⁵
C. $x_n = x_{\lfloor n/2 \rfloor} + n$ D. $x_n = x_{n-1} + x_{n-2}$

gatecse-2008 algorithms recurrence-relation normal

Answer key 

1.35.20 Recurrence Relation: GATE CSE 2008 | Question: 79



Let x_n denote the number of binary strings of length n that contain no consecutive 0s. ⁶

The value of x_5 is

- A. 5 B. 7 C. 8 D. 16 ⁷

gatecse-2008 algorithms recurrence-relation normal ⁸

Answer key 

1.35.21 Recurrence Relation: GATE CSE 2009 | Question: 35



The running time of an algorithm is represented by the following recurrence relation: ⁹

$$T(n) = \begin{cases} n & n \leq 3 \\ T(\frac{n}{3}) + cn & \text{otherwise} \end{cases} \quad ^{10}$$

Which one of the following represents the time complexity of the algorithm? ¹¹

- A. $\Theta(n)$ B. $\Theta(n \log n)$ ¹²
C. $\Theta(n^2)$ D. $\Theta(n^2 \log n)$

gatecse-2009 algorithms recurrence-relation time-complexity normal ¹³

Answer key  ¹⁴

1.35.22 Recurrence Relation: GATE CSE 2012 | Question: 16



The recurrence relation capturing the optimal execution time of the *Towers of Hanoi* problem with n discs is ¹⁵

- A. $T(n) = 2T(n-2) + 2$ B. $T(n) = 2T(n-1) + n$ ¹⁷
C. $T(n) = 2T(n/2) + 1$ D. $T(n) = 2T(n-1) + 1$

gatecse-2012 algorithms easy recurrence-relation ¹⁸

Answer key  ¹⁹

1.35.23 Recurrence Relation: GATE CSE 2014 | Set 2 | Question: 13



Which one of the following correctly determines the solution of the recurrence relation with $T(1) = 1$? ²¹

$$T(n) = 2T\left(\frac{n}{2}\right) + \log n \quad ^{22}$$

- A. $\Theta(n)$ B. $\Theta(n \log n)$ C. $\Theta(n^2)$ D. $\Theta(\log n)$ ²³

gatecse-2014-set2 algorithms recurrence-relation normal

Answer key 

1.35.24 Recurrence Relation: GATE CSE 2015 | Set 1 | Question: 49



Let a_n represent the number of bit strings of length n containing two consecutive 1s. What is the recurrence relation for a_n ?

- A. $a_{n-2} + a_{n-1} + 2^{n-2}$ B. $a_{n-2} + 2a_{n-1} + 2^{n-2}$ 2
C. $2a_{n-2} + a_{n-1} + 2^{n-2}$ D. $2a_{n-2} + 2a_{n-1} + 2^{n-2}$

gatecse-2015-set1 algorithms recurrence-relation normal 3

Answer key

1.35.25 Recurrence Relation: GATE CSE 2015 | Set 3 | Question: 39



Consider the following recursive C function. 5

```
void get(int n) 6
{
    if (n < 1) return;
    get (n-1);
    get (n-3);
    printf("%d", n);
}
```

If $get(6)$ function is being called in $main()$ then how many times will the $get()$ function be invoked before returning to the $main()$? 7

- A. 15 B. 25 C. 35 D. 45 8

gatecse-2015-set3 algorithms recurrence-relation normal 9

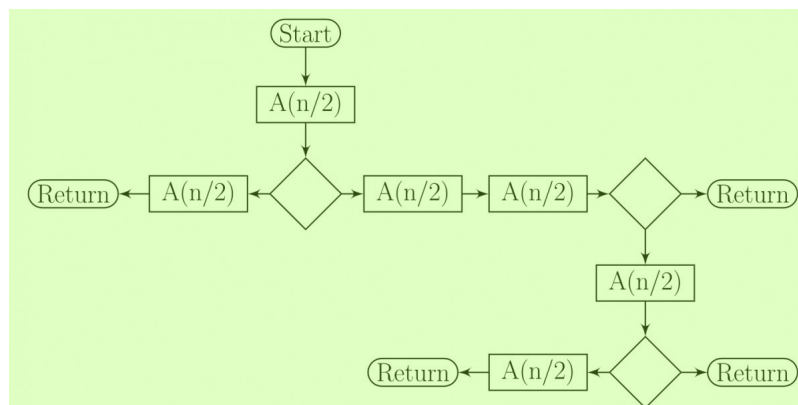
Answer key 10

1.35.26 Recurrence Relation: GATE CSE 2016 | Set 2 | Question: 39



The given diagram shows the flowchart for a recursive function $A(n)$. Assume that all statements, except for the recursive calls, have $O(1)$ time complexity. If the worst case time complexity of this function is $O(n^\alpha)$, then the least possible value (accurate up to two decimal positions) of α is _____.

Flow chart for Recursive Function $A(n)$. 12



13

gatecse-2016-set2 algorithms time-complexity recurrence-relation normal numerical-answers

Answer key

1.35.27 Recurrence Relation: GATE CSE 2017 | Set 2 | Question: 30



Consider the recurrence function

$$T(n) = \begin{cases} 2T(\sqrt{n}) + 1, & n > 2 \\ 2, & 0 < n \leq 2 \end{cases}$$

Then $T(n)$ in terms of Θ notation is

- A. $\Theta(\log \log n)$ B. $\Theta(\log n)$

C. $\Theta(\sqrt{n})$ D. $\Theta(n)$ ¹

gatecse-2017-set2 algorithms recurrence-relation

Answer key

1.35.28 Recurrence Relation: GATE CSE 2020 | Question: 2

For parameters a and b , both of which are $\omega(1)$, $T(n) = T(n^{1/a}) + 1$, and $T(b) = 1$. Then $T(n)$ is ²A. $\Theta(\log_a \log_b n)$ B. $\Theta(\log_{ab} n)$ ³C. $\Theta(\log_b \log_a n)$ D. $\Theta(\log_2 \log_2 n)$ gatecse-2020 algorithms recurrence-relation one-mark ⁴

Answer key

1.35.29 Recurrence Relation: GATE CSE 2021 | Set 1 | Question: 30

Consider the following recurrence relation. ⁵

$$T(n) = \begin{cases} T(n/2) + T(2n/5) + 7n & \text{if } n > 0 \\ 1 & \text{if } n = 0 \end{cases} \quad \sup>6$$

Which one of the following options is correct? ⁷A. $T(n) = \Theta(n^{5/2})$ B. $T(n) = \Theta(n \log n)$ ⁸C. $T(n) = \Theta(n)$ D. $T(n) = \Theta((\log n)^{5/2})$ gatecse-2021-set1 algorithms recurrence-relation time-complexity two-marks ⁹Answer key ¹⁰

1.35.30 Recurrence Relation: GATE CSE 2021 | Set 2 | Question: 39

For constants $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers: ¹¹

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n) \quad \sup>12$$

Which one of the following options is correct about the recurrence $T(n)$? ¹³A. If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\Theta(n \log_2(n))$ ¹⁴B. If $f(n)$ is $\frac{n}{\log_2(n)}$, then $T(n)$ is $\Theta(\log_2(n))$ C. If $f(n)$ is $O(n^{\log_b(a) - \epsilon})$ for some $\epsilon > 0$, then $T(n)$ is $\Theta(n^{\log_b(a)})$ D. If $f(n)$ is $\Theta(n^{\log_b(a)})$, then $T(n)$ is $\Theta(n^{\log_b(a)})$ gatecse-2021-set2 algorithms recurrence-relation two-marks ¹⁵Answer key ¹⁶

1.35.31 Recurrence Relation: GATE CSE 2024 | Set 1 | Question: 32

Consider the following recurrence relation:

$$T(n) = \begin{cases} \sqrt{n}T(\sqrt{n}) + n & \text{for } n \geq 1, \\ 1 & \text{for } n = 1 \end{cases}$$

Which one of the following options is CORRECT?

A. $T(n) = \Theta(n \log \log n)$ B. $T(n) = \Theta(n \log n)$ C. $T(n) = \Theta(n^2 \log n)$ D. $T(n) = \Theta(n^2 \log \log n)$

gatecse-2024-set1 algorithms recurrence-relation two-marks

Answer key

1.35.32 Recurrence Relation: GATE CSE 2025 | Set 1 | Question: 10



Consider the following recurrence relation: 1

$$T(n) = 2T(n-1) + n2^n \text{ for } n > 0, \quad T(0) = 1 \quad 2$$

Which ONE of the following options is CORRECT? 3

- A. $T(n) = \Theta(n^2 2^n)$ B. $T(n) = \Theta(n 2^n)$ 4
 C. $T(n) = \Theta((\log n)^2 2^n)$ D. $T(n) = \Theta(4^n)$

gatecse2025-set1 algorithms time-complexity recurrence-relation one-mark

Answer key 5

1.35.33 Recurrence Relation: GATE CSE 2026 | Set 2 | Question: 15



Which of the following can be recurrence relation(s) corresponding to an algorithm with time complexity $\Theta(n)$? 6

- A. $T(n) = T(n-1) + 1, \quad T(1) = 1$ B. $T(n) = 2T(\frac{n}{2}) + 1, \quad T(1) = 1$ 8
 C. $T(n) = 2T(\frac{n}{2}) + n, \quad T(1) = 1$ D. $T(n) = T(n-1) + n, \quad T(1) = 1$

gatecse-2026-set2 algorithms recurrence-relation multiple-selects one-mark 9

Answer key 10

1.35.34 Recurrence Relation: GATE IT 2004 | Question: 57



Consider a list of recursive algorithms and a list of recurrence relations as shown below. Each recurrence relation corresponds to exactly one algorithm and is used to derive the time complexity of the algorithm. 11

	Recursive Algorithm		Recurrence Relation
P	Binary search	I.	$T(n) = T(n-k) + T(k) + cn$
Q.	Merge sort	II.	$T(n) = 2T(n-1) + 1$
R.	Quick sort	III.	$T(n) = 2T(n/2) + cn$
S.	Tower of Hanoi	IV.	$T(n) = T(n/2) + 1$

Which of the following is the correct match between the algorithms and their recurrence relations? 12

- A. P-II, Q-III, R-IV, S-I B. P-IV, Q-III, R-I, S-II 14
 C. P-III, Q-II, R-IV, S-I D. P-IV, Q-II, R-I, S-III

gateit-2004 algorithms recurrence-relation normal match-the-following 15

Answer key 16

1.35.35 Recurrence Relation: GATE IT 2005 | Question: 51



Let $T(n)$ be a function defined by the recurrence 17

$$T(n) = 2T(n/2) + \sqrt{n} \text{ for } n \geq 2 \text{ and } T(1) = 1$$

Which of the following statements is TRUE? 18

- A. $T(n) = \Theta(\log n)$ B. $T(n) = \Theta(\sqrt{n})$ 19
 C. $T(n) = \Theta(n)$ D. $T(n) = \Theta(n \log n)$

gateit-2005 algorithms recurrence-relation easy 20

Answer key 21

1.35.36 Recurrence Relation: GATE IT 2008 | Question: 44



When $n = 2^{2k}$ for some $k \geq 0$, the recurrence relation

$$T(n) = \sqrt{(2)}T(n/2) + \sqrt{n}, \quad T(1) = 1$$

evaluates to :

A. $\sqrt{n}(\log n + 1)$
C. $\sqrt{n} \log \sqrt{n}$

B. $\sqrt{n} \log n$
D. $n \log \sqrt{n}$

gateit-2008 algorithms recurrence-relation normal 2

Answer key

1.36

Recursion (5)

 Practice Test: Test 1 (10Q)

1.36.1 Recursion: GATE CSE 1995 | Question: 2.9



A language with string manipulation facilities uses the following operations 4

head(s): first character of a string

tail(s): all but exclude the first character of a string

concat(s1, s2): s1s2

For the string "acbc" what will be the output of 6

`concat(head(s), head(tail(tail(s))))`

A. ac B. bc C. ab D. cc 8

gate1995 algorithms normal recursion 9

Answer key

1.36.2 Recursion: GATE CSE 2007 | Question: 44



In the following C function, let $n \geq m$. 10

```
int gcd(n,m) {  
    if (n%m == 0) return m;  
    n = n%m;  
    return gcd(m,n);  
}
```

How many recursive calls are made by this function? 12

A. $\Theta(\log_2 n)$ B. $\Omega(n)$ 13
C. $\Theta(\log_2 \log_2 n)$ D. $\Theta(\sqrt{n})$

gatecse-2007 algorithms recursion time-complexity normal 14

Answer key

1.36.3 Recursion: GATE CSE 2018 | Question: 45



Consider the following program written in pseudo-code. Assume that x and y are integers. 16

```
Count (x, y) {  
    if (y != 1) {  
        if (x != 1) {  
            print("x");  
            Count (x/2, y);  
        }  
        else {  
            y=y-1;  
            Count (1024, y);  
        }  
    }  
}
```

The number of times that the *print* statement is executed by the call *Count*(1024, 1024) is 18

gatecse-2018 numerical-answers algorithms recursion two-marks

Answer key

1.36.4 Recursion: GATE CSE 2021 | Set 2 | Question: 49



Consider the following ANSI C program ¹

```
#include <stdio.h>
int foo(int x, int y, int q)
{
    if ((x<=0) && (y<=0))
        return q;
    if (x<=0)
        return foo(x, y-q, q);
    if (y<=0)
        return foo(x-q, y, q);
    return foo(x, y-q, q) + foo(x-q, y, q);
}
int main( )
{
    int r = foo(15, 15, 10);
    printf("%d", r);
    return 0;
}
```

The output of the program upon execution is _____ ³

gatecse-2021-set2 algorithms recursion output numerical-answers two-marks

Answer key

1.36.5 Recursion: GATE CSE 2026 | Set 1 | Question: 51



Consider the recursive functions represented by the following code segment: ⁴

```
int bar(int n) {
    if (n == 1) return 0;
    else return 1 + bar(n/2);
}
int foo(int n) {
    if (n == 1) return 1;
    else return 1 + foo(bar(n));
}
```

The smallest positive integer n for which $f \circ o(n)$ returns 5 is _____. (answer in integer) ⁶

Note: Ignore syntax errors (if any) in the function. ⁷

gatecse-2026-set1 algorithms recursion numerical-answers two-marks ⁸

Answer key

1.37

Searching (8)

Practice Test: Test 1 (9Q) ⁹

1.37.1 Searching: GATE CSE 1996 | Question: 18



Consider the following program that attempts to locate an element x in an array $a[]$ using binary search. Assume $N > 1$. The program is erroneous. Under what conditions does the program fail? ¹¹

```
var i,j,k: integer; x: integer;
a: array[1..N] of integer;
begin i:= 1; j:= n;
repeat
    k:=(i+j) div 2;
    if a[k] < x then i:= k
    else j:= k
until (a[k] = x) or (i >= j);

if (a[k] = x) then
    writeln('x is in the array')
else
    writeln('x is not in the array')
end;
```

gate1996 algorithms searching normal descriptive

Answer key

1.37.2 Searching: GATE CSE 1996 | Question: 2.13, ISRO2016-28



The average number of key comparisons required for a successful search for sequential search on n items is

- A. $\frac{n}{2}$ B. $\frac{n-1}{2}$ C. $\frac{n+1}{2}$ D. None of the above

gate1996 algorithms easy isro2016 searching 3

Answer key

1.37.3 Searching: GATE CSE 2002 | Question: 2.10



Consider the following algorithm for searching for a given number x in an unsorted array $A[1..n]$ having n distinct values:

1. Choose an i at random from $1..n$
2. If $A[i] = x$, then Stop else Goto 1;

Assuming that x is present in A , what is the expected number of comparisons made by the algorithm before it terminates?

- A. n B. $n-1$ C. $2n$ D. $\frac{n}{2}$

gatecse-2002 searching normal 8

Answer key

1.37.4 Searching: GATE CSE 2008 | Question: 84



Consider the following C program that attempts to locate an element x in an array $Y[]$ using binary search. The program is erroneous.

```
f (int Y[10] , int x) {
    int i, j, k;
    i = 0; j = 9;
    do {
        k = (i + j) / 2;
        if ( Y[k] < x) i = k; else j = k;
    } while (Y[k] != x) && (i < j) ;
    if (Y[k] == x) printf(" x is in the array ") ;
    else printf(" x is not in the array ") ;
}
```

On which of the following contents of Y and x does the program fail?

- A. Y is [1 2 3 4 5 6 7 8 9 10] and $x < 10$
- B. Y is [1 3 5 7 9 11 13 15 17 19] and $x < 1$
- C. Y is [2 2 2 2 2 2 2 2 2 2] and $x > 2$
- D. Y is [2 4 6 8 10 12 14 16 18 20] and $2 < x < 20$ and x is even

gatecse-2008 algorithms searching normal

Answer key

1.37.5 Searching: GATE CSE 2008 | Question: 85



Consider the following C program that attempts to locate an element x in an array $Y[]$ using binary search. The program is erroneous.

```
f (int Y[10] , int x) {
    int i, j, k;
    i = 0; j = 9;
    do {
        k = (i + j) / 2;
        if ( Y[k] < x) i = k; else j = k;
    } while (Y[k] != x) && (i < j) ;
    if (Y[k] == x) printf(" x is in the array ") ;
    else printf(" x is not in the array ") ;
}
```

```
}]
```

The correction needed in the program to make it work properly is **1**

- A. Change line 6 to: if $(Y[k] < x) i = k + 1$; else $j = k - 1$; **2**
- B. Change line 6 to: if $(Y[k] < x) i = k - 1$; else $j = k + 1$;
- C. Change line 6 to: if $(Y[k] < x) i = k$; else $j = k$;
- D. Change line 7 to: } while $((Y[k] == x) \&\& (i < j))$;

gatecse-2008 algorithms searching normal **3**

Answer key

1.37.6 Searching: GATE CSE 2017 | Set 1 | Question: 48



Let A be an array of 31 numbers consisting of a sequence of 0's followed by a sequence of 1's. The problem is to find the smallest index i such that $A[i]$ is 1 by probing the minimum number of locations in A . The *worst case* number of probes performed by an *optimal* algorithm is _____.

gatecse-2017-set1 algorithms normal numerical-answers searching **5**

Answer key

1.37.7 Searching: GATE CSE 2025 | Set 2 | Question: 19



Which of the following statements regarding Breadth First Search (BFS) and Depth First Search (DFS) on an undirected simple graph G is/are TRUE? **6**

- A. A DFS tree of G is a Shortest Path tree of G .
- B. Every non-tree edge of G with respect to a DFS tree is a forward/back edge. **7**
- C. If (u, v) is a non-tree edge of G with respect to a BFS tree, then the distances from the source vertex s to u and v in the BFS tree are within ± 1 of each other.
- D. Both BFS and DFS can be used to find the connected components of G .

gatecse2025-set2 algorithms searching breadth-first-search depth-first-search multiple-selects one-mark **8**

Answer key

1.37.8 Searching: GATE CSE 2025 | Set 2 | Question: 31



An array A of length n with distinct elements is said to be bitonic if there is an index $1 \leq i \leq n$ such that $A[1..i]$ is sorted in the non-decreasing order and $A[i+1..n]$ is sorted in the non-increasing order. **9**

Which ONE of the following represents the best possible asymptotic bound for the worst-case number of comparisons by an algorithm that searches for an element in a bitonic array A ? **10**

- A. $\Theta(n)$
- B. $\Theta(1)$ **11**
- C. $\Theta(\log^2 n)$
- D. $\Theta(\log n)$

gatecse2025-set2 algorithms searching bitonic-array time-complexity two-marks **12**

Answer key **13**

1.38 Selection Sort (2)

1.38.1 Selection Sort: GATE CSE 2009 | Question: 11



What is the number of swaps required to sort n elements using selection sort, in the worst case?

- A. $\Theta(n)$
- B. $\Theta(n \log n)$
- C. $\Theta(n^2)$
- D. $\Theta(n^2 \log n)$

gatecse-2009 algorithms sorting easy selection-sort

Answer key

1.38.2 Selection Sort: GATE CSE 2013 | Question: 6



Which one of the following is the tightest upper bound that represents the number of swaps required to sort n numbers using selection sort?

- A. $O(\log n)$ B. $O(n)$ C. $O(n \log n)$ D. $O(n^2)$ ²

gatecse-2013 algorithms sorting easy selection-sort ³

Answer key ⁴

1.39

Shortest Path (8)

Practice Test: Test 1 (15Q) ⁵

1.39.1 Shortest Path: GATE CSE 2002 | Question: 12



Fill in the blanks in the following template of an algorithm to compute all pairs shortest path lengths in a directed graph G with $n * n$ adjacency matrix A . $A[i, j]$ equals 1 if there is an edge in G from i to j , and 0 otherwise. Your aim in filling in the blanks is to ensure that the algorithm is correct. ⁶

```
INITIALIZATION: For i = 1 ... n
{
  For j = 1 ... n
  {
    if a[i,j] = 0 then P[i,j] = _____ else P[i,j] = _____;
  }
}

ALGORITHM: For i = 1 ... n
{
  For j = 1 ... n
  {
    For k = 1 ... n
    {
      P[_____, _____] = min{_____, _____};
    }
  }
}
```

⁷

- Copy the complete line containing the blanks in the Initialization step and fill in the blanks. ⁸
- Copy the complete line containing the blanks in the Algorithm step and fill in the blanks.
- Fill in the blank: The running time of the Algorithm is $O(\text{_____})$.

gatecse-2002 algorithms graph-algorithms time-complexity normal descriptive shortest-path

Answer key ⁹

1.39.2 Shortest Path: GATE CSE 2003 | Question: 67



Let $G = (V, E)$ be an undirected graph with a subgraph $G_1 = (V_1, E_1)$. Weights are assigned to edges of G as follows. ¹⁰

$$w(e) = \begin{cases} 0, & \text{if } e \in E_1 \\ 1, & \text{otherwise} \end{cases}$$

A single-source shortest path algorithm is executed on the weighted graph (V, E, w) with an arbitrary vertex v_1 of V_1 as the source. Which of the following can always be inferred from the path costs computed? ¹¹

- The number of edges in the shortest paths from v_1 to all vertices of G ¹²
- G_1 is connected
- V_1 forms a clique in G
- G_1 is a tree

gatecse-2003 algorithms graph-algorithms normal shortest-path ¹³

Answer key ¹⁴

1.39.3 Shortest Path: GATE CSE 2007 | Question: 41



In an unweighted, undirected connected graph, the shortest path from a node S to every other node is computed most efficiently, in terms of *time complexity*, by

- Dijkstra's algorithm starting from S .
- Warshall's algorithm.

C. Performing a DFS starting from S .

D. Performing a BFS starting from S . 1

gatecse-2007 algorithms graph-algorithms easy shortest-path 2

Answer key

1.39.4 Shortest Path: GATE CSE 2020 | Question: 40



Let $G = (V, E)$ be a directed, weighted graph with weight function $w : E \rightarrow \mathbb{R}$. For some function $f : V \rightarrow \mathbb{R}$, for each edge $(u, v) \in E$, define $w'(u, v)$ as $w(u, v) + f(u) - f(v)$. 3

Which one of the options completes the following sentence so that it is TRUE? 4

"The shortest paths in G under w are shortest paths under w' too, _____".

- A. for every $f : V \rightarrow \mathbb{R}$
- B. if and only if $\forall u \in V$, $f(u)$ is positive
- C. if and only if $\forall u \in V$, $f(u)$ is negative
- D. if and only if $f(u)$ is the distance from s to u in the graph obtained by adding a new vertex s to G and edges of zero weight from s to every vertex of G 5

gatecse-2020 algorithms graph-algorithms two-marks shortest-path 6

Answer key

1.39.5 Shortest Path: GATE CSE 2025 | Set 1 | Question: 8



Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G . For any two vertices, u and v , let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T , respectively. Which ONE of the options is CORRECT for all possible G, T, u and v ? 7

- A. $d_1(u, v) = d_2(u, v)$
- B. $d_1(u, v) \leq d_2(u, v)$ 8
- C. $d_1(u, v) \geq d_2(u, v)$
- D. $d_1(u, v) \neq d_2(u, v)$

gatecse2025-set1 algorithms minimum-spanning-tree shortest-path one-mark 9

Answer key

1.39.6 Shortest Path: GATE CSE 2025 | Set 2 | Question: 27



Let G be an edge-weighted undirected graph with positive edge weights. Suppose a positive constant α is added to the weight of every edge. 10

Which ONE of the following statements is TRUE about the minimum spanning trees (MSTs) and shortest paths (SPs) in G before and after the edge weight update? 11

- A. Every MST remains an MST, and every SP remains an SP. 12
- B. MSTs need not remain MSTs, and every SP remains an SP.
- C. Every MST remains an MST, and SPs need not remain SPs.
- D. MSTs need not remain MSTs, and SPs need not remain SPs.

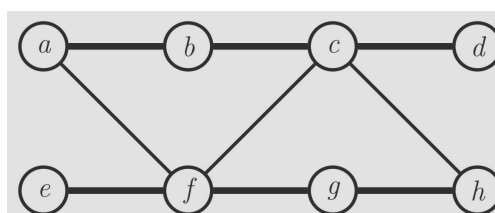
gatecse2025-set2 algorithms minimum-spanning-tree shortest-path two-marks

Answer key

1.39.7 Shortest Path: GATE DA 2025 | Question: 48



Let G be a simple, unweighted, and undirected graph. A subset of the vertices and edges of G are shown below.



It is given that $a - b - c - d$ is a shortest path between a and d ; $e - f - g - h$ is a shortest path between e and h ; $a - f - c - h$ is a shortest path between a and h . Which of the following is/are NOT the edges of G ? 1

- A. (b, d) B. (b, g) C. (b, h) D. (e, g) 2

gateda-2025 algorithms shortest-path multiple-selects two-marks 3

Answer key

1.39.8 Shortest Path: GATE IT 2007 | Question: 3, UGCNET-June2012-III: 34 4

Consider a weighted, undirected graph with positive edge weights and let uv be an edge in the graph. It is known that the shortest path from the source vertex s to u has weight 53 and the shortest path from s to v has weight 65. Which one of the following statements is always TRUE? 5

- A. Weight $(u, v) \leq 12$ B. Weight $(u, v) = 12$ 6
C. Weight $(u, v) \geq 12$ D. Weight $(u, v) > 12$

gateit-2007 algorithms graph-algorithms normal ugcnetcse-june2012-paper3 shortest-path 7

Answer key 8

1.40 Sorting (22)

Practice Tests: Test 1 (15Q) Test 2 (15Q) Test 3 (15Q) Test 4 (7Q) 9

1.40.1 Sorting: GATE CSE 1988 | Question: 1iii 10

Quicksort is _____ efficient than heapsort in the worst case. 11

gate1988 algorithms sorting fill-in-the-blanks easy 12

Answer key

1.40.2 Sorting: GATE CSE 1990 | Question: 3-v 13

The complexity of comparison based sorting algorithms is: 14

- A. $\Theta(n \log n)$ B. $\Theta(n)$ 15
C. $\Theta(n^2)$ D. $\Theta(n\sqrt{n})$

gate1990 normal algorithms sorting easy time-complexity multiple-selects

Answer key

1.40.3 Sorting: GATE CSE 1991 | Question: 01,vii 16

The minimum number of comparisons required to sort 5 elements is _____ 17

gate1991 normal algorithms sorting numerical-answers 18

Answer key

1.40.4 Sorting: GATE CSE 1991 | Question: 13 19

Give an optimal algorithm in pseudo-code for sorting a sequence of n numbers which has only k distinct numbers (k is not known a Priori). Give a brief analysis for the time-complexity of your algorithm. 20

gate1991 sorting time-complexity algorithms difficult descriptive 21

Answer key

1.40.5 Sorting: GATE CSE 1992 | Question: 02,ix 22

Following algorithm(s) can be used to sort n in the range $[1 \dots n^3]$ in $O(n)$ time 23

- a. Heap sort b. Quick sort c. Merge sort d. Radix sort 24

gate1992 easy algorithms sorting multiple-selects

Answer key

1.40.6 Sorting: GATE CSE 1996 | Question: 14



A two dimensional array $A[1..n][1..n]$ of integers is partially sorted ¹ if $\forall i, j \in [1..n-1], A[i][j] < A[i][j+1]$ and $A[i][j] < A[i+1][j]$

- a. The smallest item in the array is at $A[i][j]$ where $i = \underline{\hspace{1cm}}$ and $j = \underline{\hspace{1cm}}$. ²
- b. The smallest item is deleted. Complete the following $O(n)$ procedure to insert item x (which is guaranteed to be smaller than any item in the last row or column) still keeping A partially sorted.

```

procedure insert (x: integer);
var i, j: integer;
begin
  i:=1; j:=1, A[i][j]:=x;
  while (x >     or x >    ) do
    if A[i+1][j] < A[i][j]     then begin
      A[i][j]:=A[i+1][j]; i:=i+1;
    end
    else begin
                
    end
  end
  A[i][j]:=       
end

```

³

gate1996 algorithms sorting normal descriptive ⁴

Answer key

1.40.7 Sorting: GATE CSE 1998 | Question: 1.22



Give the correct matching for the following pairs: ⁵

(A) $O(\log n)$	(P) Selection
(B) $O(n)$	(Q) Insertion sort
(C) $O(n \log n)$	(R) Binary search
(D) $O(n^2)$	(S) Merge sort

⁶

- A. A-R B-P C-Q D-S B. A-R B-P C-S D-Q ⁷
- C. A-P B-R C-S D-Q D. A-P B-S C-R D-Q

gate1998 algorithms sorting easy match-the-following ⁸

Answer key ⁹

1.40.8 Sorting: GATE CSE 1999 | Question: 1.12



A sorting technique is called stable if ¹⁰

- A. it takes $O(n \log n)$ time ¹¹
- B. it maintains the relative order of occurrence of non-distinct elements
- C. it uses divide and conquer paradigm
- D. it takes $O(n)$ space

gate1999 algorithms sorting easy

Answer key

1.40.9 Sorting: GATE CSE 1999 | Question: 8



Let A be an $n \times n$ matrix such that the elements in each row and each column are arranged in ascending order. Draw a decision tree, which finds 1st, 2nd and 3rd smallest elements in minimum number of comparisons.

gate1999 algorithms sorting normal descriptive

Answer key